
PRISM : Decomposing Geometric Drift for Risk Diagnosis in LLM Variants

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Abstract

1 Comparing Large Language Model (LLM) variants—quantized checkpoints, fine-
2 tuned snapshots, LoRA adaptations—faces a diagnostic gap. Aggregate benchmark
3 accuracy flags degradation without explaining it, and existing representational
4 similarity metrics (e.g., CKA) reduce geometric mismatch to a single aggregate
5 score without linking to downstream performance. We propose **PRISM** (Proxy
6 Risk Inference via Structural Mapping), a framework that upper-bounds the cross-
7 entropy risk gap between a target and proxy model. PRISM combines an *exact*
8 *closed-form decomposition* of the feature alignment residual—scale mismatch
9 $\Delta\rho$ and shape mismatch $1 - \Omega$ —with a covariance-weighted head discrepancy.
10 Across four 8B-scale LLM families and five benchmarks, the bound ranks variant
11 degradation with strong Spearman correlation, and its three axes localize distinct
12 empirical settings under post-training quantization and LoRA fine-tuning, turning
13 a scalar similarity into an axis-level diagnostic. Because Ω is differentiable, the
14 shape term doubles as a training-time regularizer against catastrophic forgetting—
15 extending PRISM from a diagnostic tool into an active training signal.

16 1 Introduction

17 As large language models move from research prototypes to production deployments, a new engineer-
18 ing bottleneck emerges after pre-training. A single base checkpoint now yields many post-training
19 variants—quantized (GPTQ [6], GGUF, BitsAndBytes [3]), LoRA-adapted [11], or distilled under
20 different deployment configurations—each requiring comparison before deployment [8, 21]. Evalua-
21 tion gives only an aggregate accuracy score per benchmark, so a degraded variant is flagged but its
22 cause is not: practitioners resort to blind trial-and-error rather than targeted correction.

23 A natural shortcut is to borrow a representational similarity metric: SVCCA [22], CKA [16], and
24 related Frobenius-norm metrics each reduce two activation matrices to a single alignment score. These
25 scores, however, neither predict downstream behavior [4, 2, 15] nor identify an axis for diagnosis.
26 What is missing is a geometric quantity that lifts into a functional risk bound while keeping its
27 structural decomposition intact.

28 Constructing such a bound on the deployed model’s cross-entropy risk faces three concrete obstacles.
29 (i) No prior representational similarity—CKA, SVCCA, or generalized shape metrics—has been
30 lifted to bound the deployed model’s CE risk; the established probe-based route [9] evaluates a
31 freshly-trained linear classifier instead of the model’s own head. (ii) Cross-entropy is Lipschitz
32 in features, but the naive Lipschitz constant scales with the head’s full spectral norm and is too
33 loose to be informative over an LLM vocabulary of size $V \sim 10^5$. (iii) Post-training quantization
34 and parameter-efficient fine-tuning are studied by separate communities, with no shared geometric
35 framework that bounds the same risk gap in both settings.

Two structural properties of LLMs nonetheless make the lift tractable. Transformer LLMs factor cleanly into a non-linear backbone and a single linear prediction head (`lm_head`), with no intermediate non-linear projection between features and vocabulary logits. The Linear Representation Hypothesis [17, 18] further asserts that same-family variants learn feature manifolds related by a near-isometric linear map. Together, these properties let us lift the Procrustes feature-alignment residual into an *exact* closed-form upper bound on the cross-entropy risk gap.

We propose **PRISM** (Proxy Risk Inference via Structural Mapping; Fig. 1), a geometry-grounded, axis-level diagnostic for LLM variants. An exact decomposition of the Procrustes residual splits the feature error into a *scale mismatch* $\Delta\rho$ (activation-magnitude collapse) and a *shape mismatch* $1 - \Omega$ (feature-manifold distortion); a covariance-weighted head-discrepancy term closes the bound. Each axis is independently measurable from a single forward pass, so a PRISM failure is not merely a flag but a *direction*: scale-dominated failures localize the loss to outlier-channel corruption, shape-dominated to distorted feature geometry between tokens, and head-dominated to head perturbation with the backbone intact.

The same quantity that diagnoses a variant’s drift can also prevent it. Under frozen-head LoRA the head term vanishes and the risk gap is governed by backbone shape drift, which Ω captures differentially. Adding a weighted penalty on $1 - \Omega$ to the fine-tuning objective converts the bound from a post-hoc diagnostic into an active constraint against catastrophic forgetting (Sec. 5.4): even a weak weight suffices to reduce downstream forgetting across benchmarks.

Contributions.

1. **Theory: a closed-form CE risk bound with three diagnostic axes, from an exact scale–shape decomposition.** The Procrustes feature-alignment residual splits exactly into scale $(\Delta\rho)^2$ and shape $2\rho_T\rho_P(1-\Omega)$ arms; combined with a covariance-weighted head-discrepancy term, this yields a closed-form upper bound on the cross-entropy risk gap $|\mathcal{R}_T - \mathcal{R}_P|$ (Theorem 1; Fig. 1).
2. **Framework: a unified diagnostic that doubles as a training objective.** The bound is computed from features and head weights alone, and applies cleanly across both PTQ and frozen-head LoRA; via the differentiable trace form Ω , it directly becomes a training-time shape regularizer against catastrophic forgetting (Sec. 5.4).
3. **Empirical: rank-consistent across both settings, with axis-level localization.** Across the cross-family Llama–Qwen pair (with Ministral and DeepSeek replications in Appendix F.3) over five benchmarks, the bound ranks variants with mean Spearman $|r_s| \approx 0.82$ on PTQ and 0.835 on LoRA forgetting (*predictiveness*); its three axes localize distinct failure modes (shape distortion at Q2/Q3 PTQ, head divergence at Qwen3 Q6_K/Q8_0, scale-axis separability under cross-task LoRA drift; *decomposability*); and a mild shape regularizer reduces forgetting across downstream benchmarks (*actionability*).

2 Related Work

Representational Similarity and Procrustes Shape Metrics. Representational similarity statistics—SVCCA [22], CKA [16], and the generalized shape metrics framework [25]—measure geometric resemblance between activation matrices, with the latter’s squared Procrustes distance already carrying the nuclear norm $\|Z_T^T Z_P\|_*$ that appears in our Ω_N . The link from these summaries to downstream behavior remains an open problem [4, 2, 15]; a recent decodability bound [9] reaches downstream via whitened kernels and newly-trained linear probes rather than the deployed prediction heads we evaluate. The Linear Representation Hypothesis [17, 18] motivates restricting alignment to the orthogonal group, and the Platonic Representation Hypothesis [12] corroborates same-family shared geometry. Our formulation differs in retaining activation scale via $(\Delta\rho)^2$, modeling head mismatch through a covariance-weighted term, and lifting the Procrustes residual into a closed-form upper bound on cross-entropy risk evaluated on each model’s deployed head.

Post-Training Quantization and Its Evaluation. Post-training quantization is the dominant deployment path for LLMs under memory constraints: GPTQ [6] uses approximate second-order information for layer-wise reconstruction; GGUF and BitsAndBytes [3] are simpler round-to-nearest schemes whose degradation tracks activation outliers [26]. Existing cheap diagnostics—layer-wise reconstruction loss or weight quantization error—are indirect proxies that do not account for nonlinear error accumulation through the full network. PRISM contributes a complementary, generation-free

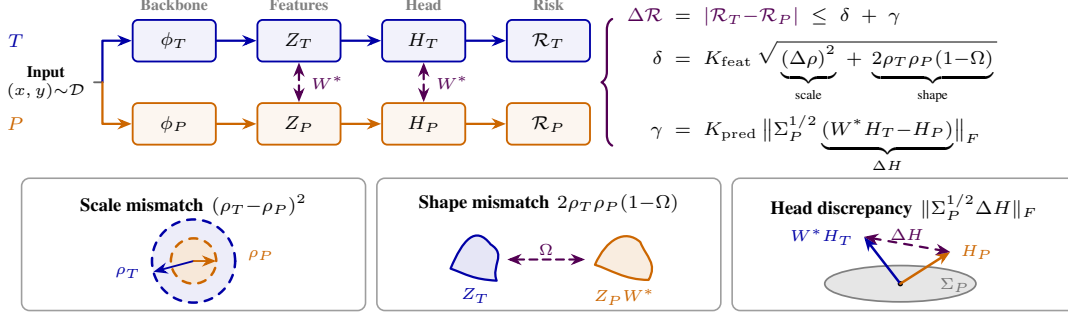


Figure 1: **PRISM decomposition of the risk gap.** The cross-entropy risk gap $|\mathcal{R}_T - \mathcal{R}_P|$ between target T and proxy P is bounded by a feature alignment error δ and a head discrepancy γ (Thm. 1). Under Procrustes alignment $W^* = \arg \min_{W \in \mathcal{O}(d)} \|Z_T - Z_P W\|_F$, δ decomposes exactly into *scale mismatch* $(\Delta \rho)^2 = (\rho_T - \rho_P)^2$ and *shape mismatch* $2\rho_T \rho_P (1 - \Omega)$ (Prop. 1); γ measures $\Delta H = W^* H_T - H_P$ re-weighted by the proxy covariance Σ_P , vanishing when $H_T = H_P$ (frozen-head LoRA). Each axis localizes a distinct empirical regime: shape distortion at low-bit PTQ, output-embedding head divergence at GGUF Q6/Q8 tiers that quantize the `lm_head` (Qwen3-Base Q6_K on MMLU in Table 11: $\delta=0.75$ vs. $\gamma=60.16$; absent under BnB/GPTQ where $\gamma \equiv 0$), and scale-axis separability under LoRA forgetting via the BBQ-vs-TruthfulQA drift contrast (Sec. 5.2).

geometric decomposition: scale collapse, structural distortion, and head divergence are each measured from a single forward pass, localizing which axis carries the degradation.

Catastrophic Forgetting under Parameter-Efficient Fine-Tuning. LoRA-based fine-tuning [11] risks catastrophic forgetting of pre-training knowledge, even when the `lm_head` stays frozen. Weight-space remedies—EWC [14] and replay-based methods—constrain parameter movement; a recent geometric analysis [24] empirically fits LoRA forgetting to *gradient*-subspace angles. Our formulation uses *hidden-state* geometry: feature-manifold drift $(\Omega, \Delta \rho)$ relative to the frozen base bounds the downstream risk gap and is usable both as a post-hoc diagnostic and as a training-time penalty (Sec. 3.5).

Efficient Evaluation and Performance Forecasting. Scaling laws [13, 10] predict aggregate loss but not variant-specific degradation; item-response-theory subsets [20, 19] preserve variant orderings without linking representational change to expected accuracy shift; weak-to-strong generalization [1] targets label generation rather than risk estimation. PRISM is geometry-grounded: the bound is computed directly from representational geometry and decomposes each variant’s degradation into three diagnostic axes.

3 A Geometric Bound on the Cross-Entropy Risk Gap

This section derives the **PRISM (Proxy Risk Inference via Structural Mapping)** bound. From two structural properties of LLMs—a linear `lm_head` on top of a non-linear backbone, and the Linear Representation Hypothesis—we obtain a closed-form upper bound on the cross-entropy risk gap $|\mathcal{R}_T - \mathcal{R}_P|$ that decomposes into three diagnostic axes: scale, shape, and head. We first state the setup (Sec. 3.1), then derive the bound (Secs. 3.2–3.3), extend it to autoregressive generation (Sec. 3.4), and lift it into a training-time regularizer (Sec. 3.5).

3.1 Setup

We consider a *Target model* T (e.g., a full-precision base) and a *Proxy model* P (e.g., a quantized or fine-tuned variant) with shared hidden dimension d . Each model factors into a Transformer backbone $\phi_M : \mathcal{X} \rightarrow \mathbb{R}^d$ followed by a linear prediction head $H_M \in \mathbb{R}^{d \times V}$ (the `lm_head` projecting onto a vocabulary of size V):

$$f_M(x) = \text{softmax}(\phi_M(x) \cdot H_M), \quad M \in \{T, P\}. \quad (1)$$

116 The risk of M under data distribution $\mathcal{D}$ is the cross-entropy expectation

$$\mathcal{R}_M = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(\phi_M(x)H_M, y)], \quad \ell(v, y) = -v_y + \log \sum_j e^{v_j}, \quad v \in \mathbb{R}^V. \quad (2)$$

117 Following the *Linear Representation Hypothesis* [17, 18], we assume the two backbones lie in a
 118 common family: there exists an orthogonal $W \in \mathcal{O}(d)$ such that $\phi_T(x) \approx \phi_P(x)W$ for all x . On a
 119 calibration sample of size n , we collect features into $Z_T, Z_P \in \mathbb{R}^{n \times d}$ and define the RMS feature
 120 scale $\rho_M = \|Z_M\|_F / \sqrt{n}$ and the empirical second-moment matrix $\Sigma_P = Z_P^\top Z_P / n$.

121 3.2 The Unified Risk Bound

122 To bound $|\mathcal{R}_T - \mathcal{R}_P|$ we introduce a hybrid risk $\mathcal{R}_{P \rightarrow T} := \mathbb{E}[\ell(\phi_P(x)W \cdot H_T, y)]$ that evaluates the
 123 target’s head on the proxy’s aligned features. The triangle inequality $|\mathcal{R}_T - \mathcal{R}_P| \leq |\mathcal{R}_T - \mathcal{R}_{P \rightarrow T}| +$
 124 $|\mathcal{R}_{P \rightarrow T} - \mathcal{R}_P|$ splits the gap into two components, each upper-bounded by a closed-form geometric
 125 quantity: a *feature alignment error* δ and a *head discrepancy* γ (full derivations in Appendix A).

126 **Feature error** δ . The cross-entropy loss is Lipschitz in the feature vector with constant K_{feat} ,
 127 derived via a simplex polarization argument in Appendix A.3. The alignment residual $\|Z_T -$
 128 $Z_P W\|_F^2 / n$ admits an exact identity for every orthogonal W :

129 **Proposition 1** (Exact Scale–Shape Decomposition). *For any $W \in \mathcal{O}(d)$,*

$$\frac{1}{n} \|Z_T - Z_P W\|_F^2 = \underbrace{(\rho_T - \rho_P)^2}_{\text{Scale Mismatch } (\Delta\rho)^2} + \underbrace{2\rho_T \rho_P (1 - \Omega_W)}_{\text{Shape Mismatch}}, \quad (3)$$

130 where Ω_W is the trace Procrustes similarity:

$$\Omega_W(Z_T, Z_P) := \frac{\text{Tr}(Z_T^\top Z_P W)}{\|Z_T\|_F \|Z_P\|_F} \in [-1, 1]. \quad (4)$$

131 Combined with the Lipschitz bound (Appendix A), we define

$$\delta := K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W)}, \quad |\mathcal{R}_T - \mathcal{R}_{P \rightarrow T}| \leq \delta. \quad (5)$$

132 **Head error** γ . Let $\Delta H := WH_T - H_P$ for the alignment W ; we define $\gamma := K_{\text{pred}} \|\Sigma_P^{1/2} \Delta H\|_F$
 133 with $K_{\text{pred}} \leq \sqrt{2}$ (Appendix A.5), so that $|\mathcal{R}_{P \rightarrow T} - \mathcal{R}_P| \leq \gamma$. The covariance weighting $\Sigma_P^{1/2}$
 134 ensures that only head misalignment within the *active subspace* of the data contributes.

135 **Theorem 1** (Unified Risk Bound). *For any $W \in \mathcal{O}(d)$,*

$$|\mathcal{R}_T - \mathcal{R}_P| \leq \mathcal{B} := \underbrace{K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W)}}_{\delta: \text{feature alignment error}} + \underbrace{K_{\text{pred}} \|\Sigma_P^{1/2} (WH_T - H_P)\|_F}_{\gamma: \text{head discrepancy}}. \quad (6)$$

136 We refer to $\mathcal{B} = \delta + \gamma$ as the PRISM bound; it is the diagnostic quantity used throughout Sec. 5.

137 3.3 Three Diagnostic Axes

138 Theorem 1 (illustrated in Fig. 1) decomposes the risk gap along three *independently measurable* axes,
 139 each attached to a distinct failure mode of the proxy.

140 **Scale mismatch** $\Delta\rho$. Divergence in activation magnitude directly amplifies the feature error.
 141 Aggressive bit-width reduction clips activation outliers [3, 26], shrinking $\rho_P < \rho_T$; the scale axis
 142 isolates this *scale collapse*.

143 **Shape mismatch** $1 - \Omega_W$. The shape term captures the distortion of the feature manifold up to
 144 rotation. A drop in Ω_W signals that the relative arrangement of token representations has been
 145 corrupted—what we term *structural distortion*.

146 **Head divergence** $\|\Sigma_P^{1/2} (WH_T - H_P)\|_F$. The head term quantifies how differently the two
 147 prediction heads interpret features, weighted by Σ_P so that only directions where the data has support
 148 can contribute.

149 **PRISM admits a family of Procrustes similarities.** The family $\{\Omega_W\}$ of Prop. 1 includes the trace
 150 form $\Omega_{W=I}$ and the nuclear form $\Omega_N := \Omega_{W=W_N}$ (Appendix B); a related Frobenius similarity Ω_F ,
 151 tied to CKA, is discussed there as an external comparison. The main text commits to $\Omega := \Omega_{W=I}$
 152 because the frozen-head settings studied here—LoRA and FP16-head PTQ—keep $H_T = H_P$ in most
 153 configurations, so the head term simplifies at $W = I$ and leaves the risk gap tractable; the $W = W_N$
 154 counterpart is reported as a Sec. 5.5 ablation.

155 3.4 Extension to Autoregressive Generation

156 The bound applies unchanged to sequence-level generation. For a sequence (c, y) with context c and
 157 target continuation $y = (y_1, \dots, y_{|y|})$, the autoregressive risk is

$$\mathcal{R}_M^{\text{AR}} = \mathbb{E}_{(c,y) \sim \mathcal{D}} \left[\frac{1}{|y|} \sum_{\tau=1}^{|y|} \ell(\phi_M(c, y_{<\tau}) \cdot H_M, y_\tau) \right]. \quad (7)$$

158 Under teacher forcing, all token-level features $\phi_M(c, y_{<\tau})$ are extracted in a single forward pass and
 159 collected into a matrix $Z_M^{\text{AR}} \in \mathbb{R}^{N \times d}$, where N is the total number of target tokens stacked across
 160 the calibration set (replacing the per-input count n of Sec. 3.1 in the AR setting); Theorem 1 then
 161 applies directly to $(Z_T^{\text{AR}}, Z_P^{\text{AR}})$, unifying point-wise classification, language modeling, short-horizon
 162 QA, and multi-step reasoning under a single bound. The full corollary is in Appendix D.

163 3.5 From Diagnostic to Training: Shape Regularization

164 LoRA fine-tuning drives the backbone shape away from its base: $\Omega(Z_0, Z_t)$ drops as training
 165 proceeds, and—because the head is frozen—the full risk gap between base θ_0 and checkpoint θ_t
 166 reduces to backbone drift alone,

$$|\mathcal{R}_0 - \mathcal{R}_t| \leq K_{\text{feat}} \sqrt{(\rho_0 - \rho_t)^2 + 2\rho_0\rho_t(1 - \Omega)}. \quad (8)$$

167 Constraining shape should therefore reduce backbone drift and suppress catastrophic forgetting on
 168 downstream tasks. Since $1 - \Omega$ is differentiable in Z_t , we augment the fine-tuning objective with a
 169 shape-preserving penalty:

$$L_{\text{total}} = \underbrace{L_{\text{CE}}(\theta_t; \mathcal{D}_{\text{FT}})}_{\text{task loss}} + \lambda (1 - \Omega(Z_0^{\text{ref}}, Z_t^{\text{ref}})), \quad (9)$$

170 where $Z_0^{\text{ref}}, Z_t^{\text{ref}}$ are base and current feature matrices on a fixed reference set (training schedule in
 171 Sec. 5.1); empirical effect is validated in Sec. 5.4.

172 4 Applications: Post-Training Quantization and LoRA Forgetting

173 The Unified Risk Bound (Theorem 1; Fig. 1) applies to two LLM lifecycle settings—post-training
 174 quantization and frozen-head LoRA fine-tuning—which differ in which term of the bound dominates
 175 the risk gap.

176 **Quantization quality estimation.** Target T is the BF16 base model and proxy P is the quantized
 177 variant. PTQ perturbs each weight independently *without applying any basis transformation*, so
 178 $W=I$ is the natural alignment and the proxy’s features stay comparable to the target’s coordinate-
 179 by-coordinate. When the protocol keeps `lm_head` in FP16 (GPTQ, BitsAndBytes), the head term γ
 180 vanishes and the bound reduces to the feature side $(\Delta\rho)^2$ and $1-\Omega$; GGUF k-quant tiers that also
 181 quantize the output embedding contribute a non-negligible γ .

182 **Geometric monitoring of catastrophic forgetting.** Target $T=\theta_0$ is the frozen base model and
 183 proxy $P=\theta_t$ is the fine-tuned checkpoint, both evaluated on downstream benchmarks *distinct* from
 184 the fine-tuning task so that $|\mathcal{R}_T - \mathcal{R}_P|$ captures catastrophic forgetting. Under standard LoRA [11]
 185 the `lm_head` stays frozen, so γ vanishes and forgetting is governed entirely by *backbone geometric*
 186 *drift* $(\Delta\rho, 1-\Omega)$; the shape regularizer of Sec. 3.5 directly penalizes the shape term $1-\Omega$.

5 Experiments

We organize the experiments around three claims: **Predictiveness**—the bound tracks $|\Delta\mathcal{R}|$ in rank order across both PTQ and LoRA variants (Sec. 5.2); **Decomposability**—its three axes localize three qualitatively distinct empirical failure modes (Sec. 5.3); and **Actionability**—the differentiable shape arm doubles as a training-time regularizer that suppresses catastrophic forgetting (Sec. 5.4). Sec. 5.5 closes with a component-wise ablation. Throughout, rank consistency is measured by Spearman $|r_s|$ between the bound and the empirical cross-entropy risk gap.

5.1 Experimental Setup

Models. The main analysis uses the cross-family pair of base models Llama-3.1-8B [8] and Qwen3-8B [21]—differing in pre-training corpus, depth, vocabulary, and tokenizer family—for both quantization and LoRA experiments. PTQ replications on Ministral-3-8B and DeepSeek-R1-Distill-Llama-8B and on the instruction-tuned counterparts of Llama, Ministral, and Qwen are in Appendix F.3; the Qwen LoRA replication is in Appendix G.

Quantization Protocols. Three PTQ families across bit-widths 2–8, spanning basic rounding to calibration-based compensation: **GGUF** (round-to-nearest, Q2_K–Q8_0), **GPTQ** [6] (4-bit, second-order reconstruction), and **BitsAndBytes** [3] (INT8, NF4, FP4).

Fine-Tuning Tasks. LoRA fine-tunes on **TruthfulQA** (factual grounding) and **BBQ** (Bias Benchmark for QA, social-context reasoning)—two tasks chosen to elicit different drift geometries (Sec. 5.3). Downstream forgetting is measured on the five standard knowledge and reasoning benchmarks below, all disjoint from either fine-tuning task.

Benchmarks and Scoring. Following Eq. (7), five benchmarks: **MMLU**, **ARC** (multiple-choice knowledge), **TriviaQA**, **SQuAD** (short-horizon QA), and **GSM8K** (multi-step reasoning); language modeling (WikiText, FineWeb-Edu) is in Appendix F.3. All risks are computed teacher-forced—prompt c and targets y scored in a single forward pass over the gold span—matching each benchmark’s scoring rule by construction and producing deterministic per-sample residuals on the same scale as the LoRA fine-tuning objective.

Calibration and Hyperparameters. PRISM and $|\Delta\mathcal{R}|$ are evaluated on fixed held-out subsets shared across all variants of a base (512 samples per benchmark for PTQ, 256 for LoRA forgetting; TruthfulQA and BBQ use their full test sets). Selection variance enters $|\Delta\mathcal{R}|$ and the bound identically across variants, so it cannot affect ranking; the orderings span 1–2 orders of magnitude in $|\Delta\mathcal{R}|$ (e.g., Q2_K vs. Q8_0). For shape regularization (Sec. 5.4) we sweep $\lambda \in \{0.0, 0.1, 0.5\}$; Ω is computed on the reference set $\mathcal{D}_{\text{ref}}$ (a fixed batch of 32 sequences from the pre-training distribution, disjoint from $\mathcal{D}_{\text{FT}}$), refreshed every $k=8$ micro-steps. Checkpoints every 25 steps, analysis at step 300. All experiments use a single NVIDIA RTX 5090 (32 GB).

5.2 Predictiveness: The Bound Tracks the Risk Gap

Quantization. Fig. 2 demonstrates PTQ predictiveness: the PRISM bound $\mathcal{B}$ tracks $|\Delta\mathcal{R}|$ with strong rank correlation across the cross-family Llama–Qwen pair (mean $|r_s| \approx 0.82$ across the 2×5 grid; the same pattern replicates on Ministral and DeepSeek in Appendix Fig. 5). Two patterns hold consistently: (i) *bit-width drives the bound monotonically*—Q8/Q6 GGUF variants sit in the low- $\mathcal{B}$ /low- $|\Delta\mathcal{R}|$ corner, while Q2 configurations move into the upper-right region where shape distortion ($1-\Omega$) dominates the feature error; (ii) *family ordering is preserved across benchmarks*—BitsAndBytes NF4/FP4 and GGUF Q3 cluster, while GPTQ-4bit (Llama, DeepSeek) sits consistently lower in $\mathcal{B}$, reflecting its second-order Hessian compensation.

Table 1 reports the per-variant decomposition for Llama-3.1-8B on MMLU and TriviaQA; the Qwen3-8B counterpart, the remaining Llama benchmarks, the feature-only (δ) scatter, parallel language-modeling results, and Ministral/DeepSeek decompositions are in Appendices F.2–F.3. As a concrete example: at the extreme Q2_K TriviaQA case, ρ_P exceeds ρ_T only mildly ($\Delta\rho \approx 1.3$), but Ω drops to 0.76, driving the shape-mismatch arm to dominate the bound—a separation invisible to any scalar similarity metric.

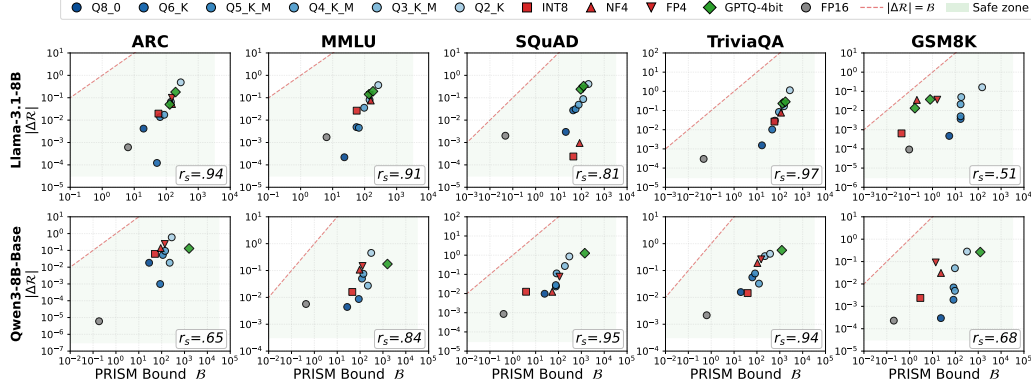


Figure 2: **The PRISM bound $\mathcal{B}$ tracks the empirical risk gap across two model families and five benchmarks.** Each subplot scatters the PRISM bound $\mathcal{B}$ (x-axis, log) against the empirical cross-entropy risk gap $|\Delta\mathcal{R}|$ (y-axis, log). Each point is one quantization variant; colors denote PTQ family (GGUF / GPTQ / BitsAndBytes). Rows: Llama-3.1-8B, Qwen3-8B. Columns: ARC, MMLU, SQuAD, TriviaQA, GSM8K. Per-subplot Spearman r_s is annotated in each panel (mean $|r_s| \approx 0.82$ across the 2×5 grid; per-component breakdown in Sec. 5.5); the red dashed $y=x$ line marks bound tightness, and the **green shaded region below it** is the safe zone where the bound provably holds. Replication on Ministral-3-8B and DeepSeek-R1-Distill-Llama-8B: Fig. 5 (Appendix F).

Table 1: Geometric decomposition for **Llama-3.1-8B** under identity alignment ($W=I$) — main task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
MMLU ($r_s=0.92$)	GGUF	FP16	138.96	138.96	0.9998	2.4695	0	6.4546	0.0017
		Q8_0	138.96	139.18	0.9988	6.7839	3.8942	23.2386	0.0002
		Q6_K	138.96	139.83	0.9945	14.6429	12.2348	55.5755	0.0049
		Q5_K_M	138.96	139.94	0.9912	18.5659	12.2455	65.8443	0.0045
		Q4_K_M	138.96	140.36	0.9764	30.3519	12.3168	96.7506	0.0356
		Q3_K_M	138.96	140.33	0.9413	47.8463	12.2598	142.3959	0.0808
	BnB	Q2_K	138.96	143.86	0.7750	94.9683	12.6318	266.0867	0.3658
		INT8	138.96	139.09	0.9880	21.5739	0	56.3886	0.0265
		NF4	138.96	144.16	0.9124	59.4744	0	155.4506	0.0750
		FP4	138.96	138.10	0.9196	55.5436	0	145.1767	0.1306
	GPTQ	GPTQ-4bit	138.96	140.37	0.9298	52.3337	0	136.7867	0.1422
TriviaQA ($r_s=0.95$)	GGUF	FP16	140.93	140.91	1.0000	0.0178	0	0.0467	0.0003
		Q8_0	140.93	140.89	0.9996	4.2226	3.9082	16.5638	0.0015
		Q6_K	140.93	141.11	0.9969	11.1777	12.1998	46.4686	0.0102
		Q5_K_M	140.93	141.18	0.9930	16.6601	12.2055	60.8066	0.0296
		Q4_K_M	140.93	141.14	0.9795	28.5309	12.2070	91.8357	0.0842
		Q3_K_M	140.93	141.39	0.9330	51.6882	12.2335	152.4004	0.1675
	BnB	Q2_K	140.93	142.25	0.7574	98.6280	12.3123	275.2004	1.1405
		INT8	140.93	141.07	0.9875	22.3310	0	58.3674	0.0266
		NF4	140.93	142.03	0.9533	43.2481	0	113.0393	0.0773
		FP4	140.93	142.54	0.9124	59.3325	0	155.0800	0.2248
	GPTQ	GPTQ-4bit	140.93	141.30	0.9347	50.9854	0	133.2628	0.2326

LoRA forgetting. The same predictiveness extends to fine-tuning-induced drift (Fig. 3): as LoRA fine-tuning proceeds on TruthfulQA or BBQ, the bound computed on *downstream* benchmarks tracks the empirical cross-entropy drift step-by-step, with mean Spearman $|r_s|=0.835$ aggregated over both fine-tuning tasks and the five downstream benchmarks, matching the strong correlation of the PTQ grid (mean $|r_s| \approx 0.82$, Fig. 2). Because the head is frozen, $\gamma=0$ and $\mathcal{B}$ reduces to backbone scale and shape drift. Crucially, the correlation holds *benchmark-by-benchmark*—each of ARC, MMLU, SQuAD, TriviaQA, GSM8K (including GSM8K, whose distribution differs substantially from the fine-tuning data)—consistent with backbone geometry being the locus of the forgetting signal across distinct evaluation families. The Qwen3-8B replication (Appendix G) reproduces both

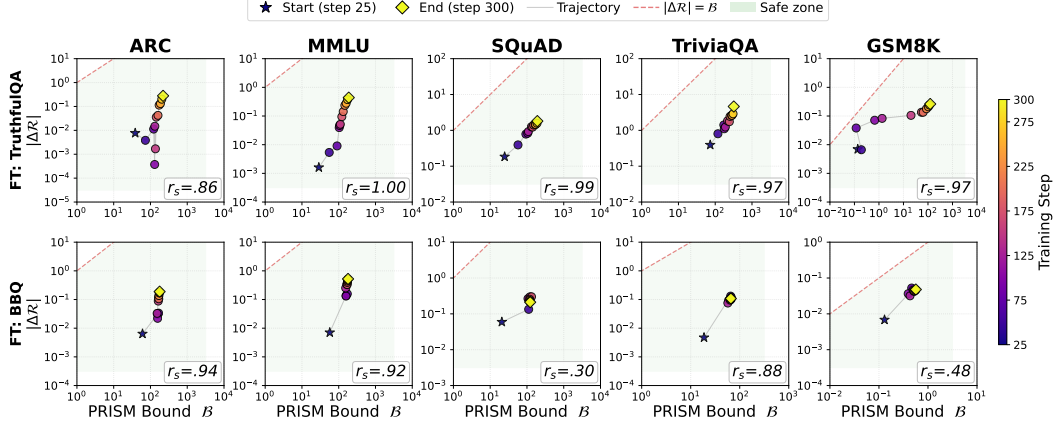


Figure 3: **Llama-3.1-8B: the PRISM bound tracks catastrophic forgetting across LoRA fine-tuning steps.** Each subplot scatters the bound B (x-axis, log) against the empirical forgetting $|\Delta\mathcal{R}|$ (y-axis, log) on a downstream benchmark, with one point per LoRA checkpoint colored by training step. Rows: fine-tuning task (TruthfulQA, BBQ). Columns: downstream benchmark (ARC, MMLU, SQuAD, TriviaQA, GSM8K). Under LoRA’s frozen $1m_head$, $\gamma=0$ so B reduces to backbone scale ($\Delta\rho$) and shape ($1-\Omega$) drift. Spearman r_s per subplot is annotated in each panel. Qwen3-8B replication: Appendix G.

patterns. Together, the PTQ and LoRA results show that a single closed-form bound transfers across two qualitatively different sources of model variation.

5.3 Decomposability: Three Axes, Three Failure Modes

A scalar correlation alone would already match prior work on representational similarity. The decomposition contributes the next layer: the dominant axis of Theorem 1 reads off directly from the per-variant numbers and corresponds to a distinct empirical failure mode. Three such failure modes recur in our experiments.

Shape distortion (low-bit PTQ). At Q2 and Q3 across all four families tested (Appendix F.3 for Ministral and DeepSeek), the shape term $2\rho_T\rho_P(1-\Omega)$ outweighs the scale term $(\Delta\rho)^2$ by two to three orders of magnitude (e.g., Llama-Q2_K on MMLU: 24 vs. $\sim 9,000$; Ministral-Q2_K on MMLU: ~ 9 vs. $\sim 11,000$)—consistent with low-bit PTQ corrupting the relational structure of the feature manifold rather than its global scale. Scale drift is measurable across the PTQ grid but does not cleanly dominate any single variant; the most scale-affected case is Qwen3-Base Q2_K on GSM8K (Table 11, Appendix F.3), where ρ_P rises from 267 to 313 alongside shape drift.

Head divergence (GGUF k-quant tiers that touch $1m_head$). A second setting appears when the protocol quantizes the output embedding. At Qwen3-Base Q6_K on MMLU (Table 11, Appendix F.3), backbone scale and shape are essentially perfect ($(\Delta\rho)^2=0.56$, $\Omega \approx 1$), so $\delta=0.75$, yet the quantized output embedding alone contributes $\gamma=60.16$ —making γ essentially the entire bound (Q8_0 shows the same pattern: $\delta=0.15$, $\gamma=19.0$). Under BnB INT8 the same Qwen3-Base keeps $\gamma\equiv 0$ by construction, leaving $\delta=13.2$ as the sole error source—a qualitatively different decomposition determined entirely by which protocol quantizes $1m_head$. The decomposition makes this protocol-level switch read off directly, identifying not just the magnitude of degradation but its dominant channel.

Scale-axis separability (LoRA cross-task contrast). The forgetting experiments reveal a third setting hidden by scalar metrics. Under TruthfulQA fine-tuning (Fig. 3 top row) we observe large shape drift with modest scale change; under BBQ fine-tuning (bottom row) both axes move. The decomposition $(\Delta\rho, 1-\Omega)$ in Eq. (8) thus exhibits the two arms as non-redundant channels: *different source tasks induce qualitatively different drift geometries*, and the scale axis is empirically separable from shape—an observation a single-number similarity would mask.

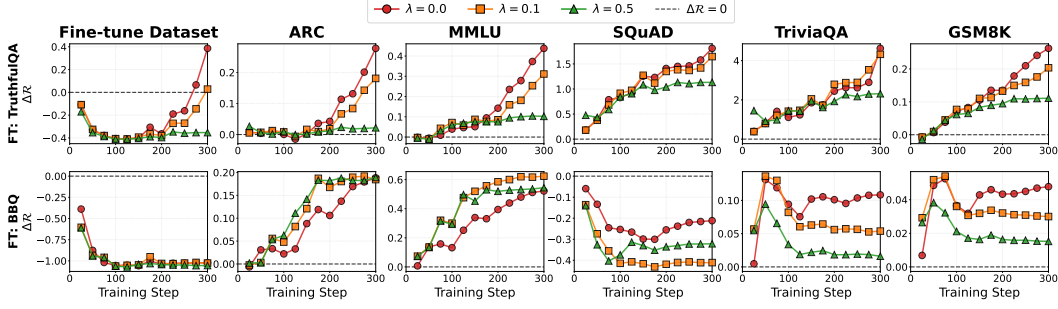


Figure 4: **PRISM-guided shape regularization on Llama-3.1-8B.** LoRA fine-tuning with the auxiliary penalty $\lambda(1 - \Omega(Z_0^{\text{ref}}, Z_t^{\text{ref}}))$ (Eq. 9) on TruthfulQA (top row) and BBQ (bottom row). At step 300: downstream forgetting $|\Delta\mathcal{R}|$ across the downstream benchmarks under $\lambda \in \{0.0, 0.1, 0.5\}$, decreasing as λ rises across benchmarks (Table 2). Qwen3-8B replication: Appendix G.

Table 2: Trace-norm forgetting decomposition for **Llama-3.1-8B** fine-tuned on **TruthfulQA** under identity alignment ($W=I$). Rows group by evaluation benchmark; each benchmark lists metrics at step 300 for $\lambda \in \{0.0, 0.1, 0.5\}$ (target = base model). Within each benchmark the row with the smallest $|\Delta\mathcal{R}|$ is marked: both $|\Delta\mathcal{R}|$ and the corresponding Ω are in bold.

Dataset	λ	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC	$\lambda=0.0$	137.55	142.67	0.8197	84.2840	0	84.2840	0.2779
	$\lambda=0.1$	137.55	140.18	0.8600	73.5186	0	192.1588	0.1810
	$\lambda=0.5$	137.55	137.13	0.9262	52.7487	0	137.8716	0.0216
MMLU	$\lambda=0.0$	138.91	143.23	0.8711	71.7461	0	71.7461	0.4380
	$\lambda=0.1$	138.91	141.57	0.8995	62.9385	0	164.5051	0.3114
	$\lambda=0.5$	138.91	138.95	0.9509	43.5242	0	113.7611	0.1023
SQuAD	$\lambda=0.0$	138.20	142.66	0.8644	73.2518	0	73.2518	1.8106
	$\lambda=0.1$	138.20	141.09	0.8903	65.4575	0	171.0891	1.6466
	$\lambda=0.5$	138.20	139.26	0.9423	47.1254	0	123.1736	1.1317
TriviaQA	$\lambda=0.0$	140.93	148.81	0.6451	122.2581	0	122.2581	4.6261
	$\lambda=0.1$	140.93	148.12	0.6731	117.0373	0	305.9053	4.3243
	$\lambda=0.5$	140.93	145.65	0.7985	91.0677	0	238.0275	2.3121
GSM8K	$\lambda=0.0$	137.15	140.24	0.9426	47.0717	0	47.0717	0.2626
	$\lambda=0.1$	137.15	138.79	0.9688	34.4920	0	90.1531	0.2034
	$\lambda=0.5$	137.15	137.77	1.0000	0.6216	0	1.6247	0.1110

273 **From diagnosis to remediation.** Each dominant axis suggests a different remediation—per-channel
274 outlier smoothing [26] for scale collapse, Hessian-aware reconstruction for shape distortion, and
275 FP16-1m_head retention for head divergence. The bound thus turns from a scalar score into an axis-
276 level diagnostic with actionable structure. We turn the differentiable shape arm into a training-time
277 regularizer in Sec. 5.4.

278 5.4 Actionability: From Diagnostic to Training Objective

279 The decomposition of Sec. 5.3 suggests an immediate intervention: since shape drift dominates LoRA
280 forgetting and $1 - \Omega$ is differentiable in Z_t , penalizing it during fine-tuning should preserve down-
281 stream task knowledge. Fig. 4 and Table 2 answer the question raised by Sec. 3.5: *does minimizing*
282 *(1 - Ω) during fine-tuning actually suppress forgetting?* For Llama-3.1-8B fine-tuned on TruthfulQA,
283 Table 2 reports the decomposition at step 300: as λ rises from 0.0 to 0.5, Ω on downstream bench-
284 marks increases monotonically (e.g., ARC 0.82→0.86→0.93, MMLU 0.87→0.90→0.95, GSM8K
285 0.94→0.97→1.00), and downstream $|\Delta\mathcal{R}|$ drops on every benchmark.

286 Three patterns emerge from Fig. 4: (i) *the reduction is broadly consistent across benchmarks*—ARC,
287 MMLU, SQuAD, TriviaQA, GSM8K move in the same direction under the same λ in essentially

Table 3: **Component-wise ablation under both alignment conventions.** Top block: identity-alignment form ($W=I$, used throughout the main text). Bottom block: Procrustes-optimal $W=W_N$ form. Same per-cell Spearman protocol throughout. Within each block, rows add bound components cumulatively (shape $\rightarrow$ + scale $\rightarrow$ + head); larger $|r_s|$ = better variant ranking; bold = best in column *within its block*. The $W=W_N$ block ranks modestly better on the full bound, as expected since W_N minimizes the alignment residual; the main text adopts the identity alignment $W=I$ for autograd compatibility and for the head-term simplification $H_T = H_P$ that holds in our frozen-1m_head regimes.

Metric	Mean $ r_s $ (10 cells)	Llama (5 benchmarks)	Qwen3 (5 benchmarks)	Wins / 10
Ω (shape only; baseline)	.804	.825	.783	2
δ (+ scale; no head)	.868	.881	.855	5
$\mathcal{B}$ (+ head; PRISM, $W=I$)	.820	.828	.813	4
Ω_N (shape only; baseline)	.806	.839	.773	4
δ_N (+ scale; no head)	.873	.898	.847	5
$\mathcal{B}_N$ (+ head; PRISM, $W=W_N$)	.912	.927	.896	8

all cases, suggesting the regularizer acts on a benchmark-agnostic backbone property rather than overfitting any single evaluation; (ii) *the trade-off is monotone and controllable*—increasing λ strengthens downstream preservation predictably, the behavior a bound-motivated regularizer should exhibit; (iii) *even the smallest setting $\lambda=0.1$ already eliminates a substantial fraction of forgetting*, indicating that a weak geometric prior is enough to curb backbone drift. The Qwen3-8B replication and BBQ per-variant tables are in Appendix G.

5.5 Ablation: Component-Wise Contributions

Table 3 measures the marginal ranking contribution of each component of the PRISM bound by progressively adding them. Under the identity-alignment form ($W=I$, top block), the shape-only similarity Ω already achieves mean $|r_s|=0.80$, indicating that the relational geometry of token features alone carries most of the variant-ranking signal. Adding the scale term to form the feature alignment error δ lifts the mean by +0.064 to 0.87 and wins the plurality of cells, making scale collapse the single largest per-component contributor. Adding the covariance-weighted head term to form the full bound $\mathcal{B}$ brings the aggregate mean slightly down to 0.82: because our suite mixes protocols that quantize the output embedding ($\gamma>0$ for GGUF k-quant tiers) with protocols that keep it in FP16 ($\gamma\equiv 0$ for GPTQ and BnB), the γ term injects non-monotone variance across the pooled scatter. Two observations keep $\mathcal{B}$ the right *default* metric. First, it is the certified upper bound of Theorem 1 and is strictly required whenever $\gamma>0$; δ alone is not a valid bound in that setting. Second, $\mathcal{B}$ has the smallest Llama–Qwen gap (≈ 0.015 vs. 0.026 and 0.042 for δ and Ω), so its ranking performance is the most robust across the two families tested.

Under the Procrustes-optimal alignment $W=W_N$ (bottom block), the cumulative pattern reverses cleanly: $\Omega_N \rightarrow \delta_N \rightarrow \mathcal{B}_N$ all increase monotonically, and the full bound $\mathcal{B}_N$ attains the strongest aggregate ranking ($|r_s|=0.91$, 8/10 cells). The head term thus benefits the bound *when the alignment can absorb the rotation between H_T and H_P* ; the non-monotonicity at $W=I$ traces to the alignment, not the bound design. The two blocks together support Theorem 1 at W_N and quantify the trade-off at $W=I$: the trace specialization sacrifices ~ 0.09 Spearman for autograd compatibility (no SVD per step), the head-term simplification under frozen 1m_head, and consistency with the regularizer of Sec. 3.5. We continue to recommend $\mathcal{B}$ as our default and $\mathcal{B}_N$ as a post-hoc reference when the additional SVD cost is acceptable.

6 Conclusion and Future Work

We presented PRISM, a geometric framework with three contributions. **Theory:** the Orthogonal Procrustes residual splits exactly into scale $(\Delta\rho)^2$ and shape $2\rho_T\rho_P(1-\Omega)$ arms; combined with a covariance-weighted head term, this yields a closed-form upper bound on the cross-entropy risk gap $|\mathcal{R}_T - \mathcal{R}_P|$ along three independently measurable axes (Theorem 1). **Framework:** the bound is computed from features and head weights alone, and applies across both PTQ and frozen-head LoRA;

via the differentiable trace form Ω , it doubles as a training-time regularizer against catastrophic forgetting. **Empirical:** rank-consistent across both settings (mean Spearman $|r_s| \approx 0.82$ on PTQ, 0.835 on LoRA forgetting), the three axes localize distinct failure modes—shape distortion at Q2/Q3 PTQ, head divergence at Qwen3 Q6_K/Q8_0, scale-axis separability under cross-task LoRA drift—and the shape regularizer reduces forgetting across downstream benchmarks.

Scope and limitations. PRISM is a *relative* metric: it bounds the cross-entropy risk gap to a reference base, not the variant’s absolute task quality. The bound is calibrated for ranking; tight estimation of $|\Delta\mathcal{R}|$ is a separate question. The shape regularizer of Sec. 5.4 provides one direct training-time intervention against the shape axis; protocol-level mitigations for the scale and head axes (per-channel outlier smoothing, Hessian-aware reconstruction, FP16-1m_head retention) are a focused empirical follow-up the diagnostic enables.

Future Work.

- **Beyond LoRA forgetting.** Extending the PRISM shape regularizer to full SFT and distillation—where backbone drift is substantially larger—would test the limits of shape anchoring as an active intervention.
- **Diagnostic applications.** The PRISM bound is computed from features and head weights alone and is therefore label-free, enabling three near-term uses: *per-sample OOD detection* via scale and shape residuals on individual inputs; *cross-scale hyperparameter transfer* using the bound between a small proxy and its scaled target to predict tuning robustness; and *production drift monitoring* on live traffic, where no labels are required.
- **Beyond LLMs.** Vision Transformers and contrastive image encoders share LLMs’ backbone-then-linear-head structure, with final-layer LayerNorm (and strict L2 normalization in CLIP/SigLIP) enforcing tight feature scales—making the PRISM scale axis naturally small. Natural targets include ViT quantization, vision-encoder distillation, and cross-modal alignment (consistent with the Platonic Representation Hypothesis [12]).

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A Proof of the Unified Risk Bound (Theorem 1)

We provide the complete proof of Theorem 1, which the main text states for general $W \in \mathcal{O}(d)$ (trace family Ω_W , Eq. 4). The proof proceeds through five self-contained steps: risk decomposition via triangle inequality, Lipschitz analysis of cross-entropy with respect to features, the alignment-residual decomposition (whose Procrustes-optimal specialization $W = W_N$ yields the nuclear form Ω_N via SVD), second-moment-adjusted head bound, and final assembly. The identity-alignment specialization $W = I$ (used throughout Sec. 5 and the regularizer of Sec. 3.5) and the $W = W_N$ specialization (used in the Sec. 5.5 ablation) follow as corollaries of the same proof.

A.1 Relation to Classical Procrustes Shape Metrics

The left-hand side of Eq. (3) at $W = W_N$ is the squared Procrustes size-and-shape distance $d_1^2(Z_T, Z_P)$ studied in the generalized shape metrics framework [25] and its recent decodability analysis [9]. Expanding d_1^2 reveals that the nuclear norm $\|Z_T^\top Z_P\|_*$ appears inside any Procrustes-based similarity; accordingly, our contribution is *not* the Procrustes distance itself. Rather, it is (a) the explicit split of this residual into the scale term $(\rho_T - \rho_P)^2$ and the shape term $2\rho_T\rho_P(1 - \Omega_N)$, which the shape-metric literature treats jointly as a single descriptive dissimilarity, and (b) its lifting to a functional-risk bound (Theorem 1). Sections 4 and 4 show that the two arms of this split have qualitatively different practical meanings and dominate the risk gap under different LLM lifecycle settings.

A.2 Step 1: Risk Decomposition via Triangle Inequality

Setup. Recall that the risk of model M is $\mathcal{R}_M = \mathbb{E}_{(x,y)}[\ell(\phi_M(x) \cdot H_M, y)]$, where ℓ is the cross-entropy loss (Eq. 2). For any orthogonal alignment $W \in \mathcal{O}(d)$, introduce the hybrid risk:

$$\mathcal{R}_{P \rightarrow T} := \mathbb{E}_{(x,y)}[\ell(\phi_P(x) W \cdot H_T, y)]. \quad (10)$$

This evaluates the target’s head H_T on proxy features aligned by W , serving as a bridge between $\mathcal{R}_T$ and $\mathcal{R}_P$. The proof below holds for any $W \in \mathcal{O}(d)$; the two specializations relevant to the paper are $W = I$ (identity alignment, trace form Ω , used throughout the main text) and $W = W_N := \arg \min_W \|Z_T - Z_P W\|_F^2$ (Procrustes-optimal, nuclear form Ω_N , Sec. 5.5 ablation).

Decomposition. By the triangle inequality on absolute values:

$$|\mathcal{R}_T - \mathcal{R}_P| \leq \underbrace{|\mathcal{R}_T - \mathcal{R}_{P \rightarrow T}|}_{\text{Feature Alignment Error } \delta} + \underbrace{|\mathcal{R}_{P \rightarrow T} - \mathcal{R}_P|}_{\text{Head Discrepancy } \gamma}. \quad (11)$$

δ measures how features differ through the same head H_T ; γ measures how heads differ on the same aligned features.

A.3 Step 2: Bounding the Feature Alignment Error δ

Lipschitz Property of Cross-Entropy with Respect to Features. We bound δ by showing that the cross-entropy loss is Lipschitz continuous in the feature vector z . Define the end-to-end composition $g_T(z, y) := \ell(z \cdot H_T, y)$, treating the classification head H_T and the cross-entropy loss as a single map from features to scalar loss. We bound g_T via direct gradient analysis rather than the Lipschitz composition rule (loss-Lipschitz $\sqrt{2}$ times head operator norm $\|H_T\|_2$): the composition rule is agnostic to the softmax output and gives the spectral constant $\sqrt{2}\|H_T\|_2$ (Remark below), whereas the gradient $\nabla_z g_T = H_T(\hat{p} - e_y)$ exposes the predicted distribution $\hat{p}$ and lets us derive a constant set by pairwise token-embedding distances instead. Our goal is to find the tightest constant K_{feat} such that:

$$|g_T(z_1, y) - g_T(z_2, y)| \leq K_{\text{feat}} \|z_1 - z_2\|_2, \quad \forall z_1, z_2, y. \quad (12)$$

Computing the Gradient. The logit vector is $v = z \cdot H_T \in \mathbb{R}^V$, and the cross-entropy loss is $\ell(v, y) = -v_y + \log \sum_{j=1}^V e^{v_j}$. The gradient with respect to the logits is:

$$\frac{\partial \ell}{\partial v} = \hat{p} - e_y, \quad (13)$$

where $\hat{p} = \text{softmax}(v) \in \mathbb{R}^V$ and e_y is the one-hot vector at index y . Applying the chain rule to obtain the gradient with respect to features:

$$\nabla_z \ell = H_T(\hat{p} - e_y) = \sum_{j=1}^V (\hat{p}_j - \delta_{jy}) h_{T,j}, \quad (14)$$

where $h_{T,j}$ denotes the j -th column of H_T (the embedding vector for token j).

Simplex Polarization. We now derive a tight bound on $\|\nabla_z \ell\|_2$ by exploiting the probability simplex constraint $\sum_j \hat{p}_j = 1$. Expanding the gradient for true class y :

$$\nabla_z \ell = (\hat{p}_y - 1) h_{T,y} + \sum_{j \neq y} \hat{p}_j h_{T,j}. \quad (15)$$

Since $1 - \hat{p}_y = \sum_{j \neq y} \hat{p}_j$, we can rewrite:

$$\nabla_z \ell = -\left(\sum_{j \neq y} \hat{p}_j\right) h_{T,y} + \sum_{j \neq y} \hat{p}_j h_{T,j} = \sum_{j \neq y} \hat{p}_j (h_{T,j} - h_{T,y}). \quad (16)$$

This reveals that the gradient is a *non-negative linear combination* of pairwise differences between incorrect-class embeddings and the correct-class embedding, with weights $\hat{p}_j$ summing to $1 - \hat{p}_y \leq 1$. Taking the norm:

$$\|\nabla_z \ell\|_2 \leq \sum_{j \neq y} \hat{p}_j \|h_{T,j} - h_{T,y}\|_2 \leq \max_{j \neq y} \|h_{T,j} - h_{T,y}\|_2 \cdot \underbrace{\sum_{j \neq y} \hat{p}_j}_{\leq 1}. \quad (17)$$

Taking the maximum over all possible true classes y :

$$K_{\text{feat}} = \max_{j,k} \|h_{T,j} - h_{T,k}\|_2. \quad (18)$$

Remark. This bound depends on the *relative distances* between token embeddings, not their absolute magnitudes. A uniform shift $H_T \rightarrow H_T + c\mathbf{1}^\top$ does not change K_{feat} . A naive Cauchy–Schwarz bound gives $K_{\text{feat}}^{\text{naive}} = \sqrt{2}\|H_T\|_2$, which is substantially looser.

From Gradient Bound to Lipschitz Inequality. Since $g_T(\cdot, y)$ is continuously differentiable, the gradient bound $\|\nabla_z g_T(z, y)\|_2 \leq K_{\text{feat}}$ implies the Lipschitz inequality stated above by the mean value theorem: for any z_1, z_2 ,

$$g_T(z_2, y) - g_T(z_1, y) = \int_0^1 \nabla_z g_T(z_1 + t(z_2 - z_1), y)^\top (z_2 - z_1) dt,$$

and Cauchy–Schwarz gives $|g_T(z_2, y) - g_T(z_1, y)| \leq K_{\text{feat}} \|z_2 - z_1\|_2$.

Applying the Lipschitz Bound. Using the Lipschitz property with $z_1 = \phi_T(x)$ and $z_2 = \phi_P(x)W$:

$$\begin{aligned} \delta &= |\mathcal{R}_T - \mathcal{R}_{P \rightarrow T}| = \left| \mathbb{E}[g_T(\phi_T(x), y) - g_T(\phi_P(x)W, y)] \right| \\ &\leq \mathbb{E}[|g_T(\phi_T(x), y) - g_T(\phi_P(x)W, y)|] \\ &\leq K_{\text{feat}} \cdot \mathbb{E}_x[\|\phi_T(x) - \phi_P(x)W\|_2]. \end{aligned} \quad (19)$$

Applying Jensen’s inequality ($\mathbb{E}[\|X\|] \leq \sqrt{\mathbb{E}[\|X\|^2]}$) to connect this to the alignment residual:

$$\delta \leq K_{\text{feat}} \cdot \sqrt{\mathbb{E}_x[\|\phi_T(x) - \phi_P(x)W\|_2^2]} = K_{\text{feat}} \cdot \sqrt{\frac{1}{n} \|Z_T - Z_P W\|_F^2}. \quad (20)$$

A.4 Step 3: Exact Geometric Decomposition

We derive the exact identity linking the alignment residual to scale and shape mismatches. The identity holds for every $W \in \mathcal{O}(d)$; we then specialize to the two cases relevant to this paper.

513 **General Identity.** For any $W \in \mathcal{O}(d)$, expand the squared Frobenius norm:

$$\begin{aligned}\|Z_T - Z_P W\|_F^2 &= \text{Tr}[(Z_T - Z_P W)^\top (Z_T - Z_P W)] \\ &= \|Z_T\|_F^2 + \|Z_P W\|_F^2 - 2 \text{Tr}(Z_T^\top Z_P W).\end{aligned}\quad (21)$$

514 Since W is orthogonal, $\|Z_P W\|_F^2 = \text{Tr}(W^\top Z_P^\top Z_P W) = \text{Tr}(Z_P^\top Z_P) = \|Z_P\|_F^2$ (cyclic trace +
515 $W^\top W = I_d$). Therefore:

$$\|Z_T - Z_P W\|_F^2 = \|Z_T\|_F^2 + \|Z_P\|_F^2 - 2 \text{Tr}(Z_T^\top Z_P W). \quad (22)$$

516 Recall $\rho_M = \|Z_M\|_F / \sqrt{n}$ (so $\|Z_M\|_F^2 / n = \rho_M^2$) and $\Omega_W := \text{Tr}(Z_T^\top Z_P W) / (\|Z_T\|_F \|Z_P\|_F)$
517 (main-text Eq. 4). Dividing Eq. (22) by n :

$$\frac{1}{n} \|Z_T - Z_P W\|_F^2 = \rho_T^2 + \rho_P^2 - 2 \rho_T \rho_P \Omega_W. \quad (23)$$

518 Adding and subtracting $2\rho_T \rho_P$ completes the square:

$$\frac{1}{n} \|Z_T - Z_P W\|_F^2 = \underbrace{(\rho_T - \rho_P)^2}_{\text{Scale Mismatch}} + \underbrace{2\rho_T \rho_P (1 - \Omega_W)}_{\text{Shape Mismatch}}. \quad (24)$$

519 This is Proposition 1 in its general form. $\square$

520 **Specialization 1: $W = I$ (identity alignment, main text).** Setting $W = I$ gives the trace form
521 $\Omega \equiv \Omega_{W=I} = \text{Tr}(Z_T^\top Z_P) / (\|Z_T\|_F \|Z_P\|_F)$, and Eq. (24) becomes

$$\frac{1}{n} \|Z_T - Z_P\|_F^2 = (\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega), \quad (25)$$

522 which is the form used throughout the main-text experiments and the regularizer of Sec. 3.5.

523 **Specialization 2: $W = W_N$ (Procrustes-optimal, ablation).** The alignment that minimizes the
524 residual is the classical orthogonal Procrustes problem [23, 7]:

$$W_N := \arg \min_{W \in \mathcal{O}(d)} \|Z_T - Z_P W\|_F^2 = \arg \max_{W \in \mathcal{O}(d)} \text{Tr}(Z_T^\top Z_P W), \quad (26)$$

525 where the second equality follows from Eq. (22) since the first two terms are W -independent. Its
526 closed-form SVD solution is well known; we restate it here for completeness. Let $Z_T^\top Z_P = U \Sigma V^\top$
527 be the SVD with $U, V \in \mathcal{O}(d)$ and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d)$, $\sigma_i \geq 0$. By cyclic trace:

$$\text{Tr}(Z_T^\top Z_P W) = \text{Tr}(\Sigma V^\top W U) = \text{Tr}(\Sigma R), \quad R := V^\top W U \in \mathcal{O}(d). \quad (27)$$

528 Since $\sigma_i \geq 0$ and $|R_{ii}| \leq 1$ for any orthogonal R , $\text{Tr}(\Sigma R) = \sum_i \sigma_i R_{ii}$ is maximized at $R = I_d$,
529 giving:

$$W_N = V U^\top, \quad \max_W \text{Tr}(Z_T^\top Z_P W) = \sum_i \sigma_i = \|Z_T^\top Z_P\|_*. \quad (28)$$

530 Therefore $\Omega_{W_N} = \Omega_N := \|Z_T^\top Z_P\|_* / (\|Z_T\|_F \|Z_P\|_F)$ (nuclear form, Appendix B Eq. 38), and
531 Eq. (24) at $W = W_N$ becomes

$$\min_{W \in \mathcal{O}(d)} \frac{1}{n} \|Z_T - Z_P W\|_F^2 = (\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_N), \quad (29)$$

532 the form underlying the Sec. 5.5 ablation. Note that W_N minimizes only the feature alignment
533 residual $\delta(W)$, not the full bound $\delta(W) + \gamma(W)$, since the head term also depends on W via
534 $W H_T - H_P$. Appendix C.3 treats the joint optimization in full and shows that W_N remains the
535 operative choice in the frozen-head LoRA and FP16-head PTQ settings studied here, where γ either
536 vanishes or is W -independent.

537 **Assembling the Feature Bound.** Combining Eq. (20) with the general decomposition Eq. (24):

$$\delta \leq K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W)}. \quad (30)$$

538 At $W = I$, replace Ω_W with Ω ; at $W = W_N$, with Ω_N .

539 A.5 Step 4: Bounding the Head Discrepancy γ

540 **Lipschitz Property with Respect to Logits.** The head discrepancy measures how much predictions
 541 differ when two different “effective heads” (WH_T vs. H_P) operate on the same proxy features. Let
 542 $z = \phi_P(x)$ denote the (unaligned) proxy feature. Then:

$$\gamma = |\mathcal{R}_{P \rightarrow T} - \mathcal{R}_P| = \left| \mathbb{E}[\ell(z \cdot WH_T, y) - \ell(z \cdot H_P, y)] \right|. \quad (31)$$

543 To bound this, we use the Lipschitz property of cross-entropy with respect to *logits*. The gradient of ℓ
 544 with respect to v is $\nabla_v \ell = \hat{p} - e_y$, with norm:

$$\|\nabla_v \ell\|_2 = \|\hat{p} - e_y\|_2. \quad (32)$$

545 Since $\hat{p}$ lies on the probability simplex, $\|\hat{p}\|_2^2 \leq \|\hat{p}\|_1^2 = 1$, and therefore $\|\hat{p} - e_y\|_2^2 = \|\hat{p}\|_2^2 -$
 546 $2\hat{p}_y + 1 \leq 2 - 2\hat{p}_y \leq 2$. The supremum $\sqrt{2}$ is approached as the model’s predicted distribution
 547 concentrates on an incorrect class ($\hat{p}_y \rightarrow 0$). Thus $K_{\text{pred}} \leq \sqrt{2}$, yielding the Lipschitz inequality
 548 $|\ell(v_1, y) - \ell(v_2, y)| \leq K_{\text{pred}} \|v_1 - v_2\|_2$ on logits.

549 **Deriving the Second-Moment Projection.** Define the head misalignment matrix $\Delta H := WH_T -$
 550 $H_P \in \mathbb{R}^{d \times V}$. The logit difference for a single sample is $z \cdot \Delta H = \phi_P(x)(WH_T - H_P)$, and:

$$\begin{aligned} \gamma &\leq \mathbb{E}[|\ell(z \cdot WH_T, y) - \ell(z \cdot H_P, y)|] \\ &\leq K_{\text{pred}} \cdot \mathbb{E}_z[\|z \cdot \Delta H\|_2]. \end{aligned} \quad (33)$$

551 Applying Jensen’s inequality on the concave square root:

$$(\mathbb{E}[\|z \cdot \Delta H\|_2])^2 \leq \mathbb{E}[\|z \cdot \Delta H\|_2^2]. \quad (34)$$

552 Expanding the squared norm using the trace operator:

$$\begin{aligned} \mathbb{E}[\|z \cdot \Delta H\|_2^2] &= \mathbb{E}[\text{Tr}(\Delta H^\top z^\top z \Delta H)] \\ &= \text{Tr}(\Delta H^\top \underbrace{\mathbb{E}[z^\top z]}_{\Sigma_P} \Delta H) \\ &= \|\Sigma_P^{1/2} \Delta H\|_F^2. \end{aligned} \quad (35)$$

553 Taking the square root:

$$\gamma \leq K_{\text{pred}} \cdot \|\Sigma_P^{1/2} (WH_T - H_P)\|_F. \quad (36)$$

554 **Interpretation.** The second-moment weighting $\Sigma_P^{1/2}$ projects the head error onto the *active sub-*
 555 *space* of the data. If the heads disagree only in directions orthogonal to the feature distribution (the
 556 null space of Σ_P), this disagreement has zero cost—it does not affect any prediction. This is strictly
 557 tighter than the spectral bound $\gamma \leq K_{\text{pred}} \|WH_T - H_P\|_2 \cdot \mathbb{E}[\|z\|_2]$, which assumes worst-case
 558 alignment between features and head error.

559 A.6 Step 5: Assembling the Unified Bound

560 For any $W \in \mathcal{O}(d)$, combining the risk decomposition (Eq. 11), the feature bound (Eq. 30), and the
 561 head bound (Eq. 36):

$$|\mathcal{R}_T - \mathcal{R}_P| \leq \delta + \gamma \leq K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W)} + K_{\text{pred}} \cdot \|\Sigma_P^{1/2} (WH_T - H_P)\|_F. \quad (37)$$

562 This is Theorem 1 in its general form. The two specializations used in the paper:

- 563 • $W = I$ (**identity alignment, main text**): $\Omega_W = \Omega$ (trace form) and $WH_T - H_P =$
 564 $H_T - H_P$, giving the bound used throughout Sec. 5 and the regularizer of Sec. 3.5.
- 565 • $W = W_N$ (**Procrustes-optimal, ablation**): $\Omega_W = \Omega_N$ (nuclear form, achieves the tightest
 566 feature-side specialization), giving the bound underlying the Sec. 5.5 ablation.

567 This completes the proof of Theorem 1. $\square$

B Tightness of Nuclear Norm over Frobenius Norm

We define the nuclear-form and Frobenius-form Procrustes similarities, show that the former yields a strictly tighter alignment residual than the latter, and clarify the relationship to CKA.

Definitions. The *nuclear-form* Procrustes similarity is

$$\Omega_N(Z_T, Z_P) := \frac{\|Z_T^\top Z_P\|_*}{\|Z_T\|_F \|Z_P\|_F} = \max_{W \in \mathcal{O}(d)} \frac{\text{Tr}(Z_T^\top Z_P W)}{\|Z_T\|_F \|Z_P\|_F}, \quad (38)$$

the maximum of the trace family Ω_W (Sec. 3.2) over orthogonal alignments, attained at the Procrustes-optimal W_N (Appendix A). The *Frobenius-form* similarity is $\Omega_F := \|Z_T^\top Z_P\|_F / (\|Z_T\|_F \|Z_P\|_F)$; unlike Ω_W , it does not arise from the alignment residual at any W and is treated here as an external comparison object connecting to CKA.

Statement. For any feature matrices Z_T, Z_P :

$$\Omega_F \leq \Omega_N, \quad (39)$$

with equality if and only if the cross-moment matrix $Z_T^\top Z_P$ has at most one nonzero singular value (i.e., $\text{rank} \leq 1$).

Proof. Let $\sigma_1, \dots, \sigma_d$ be the singular values of $Z_T^\top Z_P$. Then:

$$\|Z_T^\top Z_P\|_F = \sqrt{\sum_i \sigma_i^2}, \quad \|Z_T^\top Z_P\|_* = \sum_i \sigma_i. \quad (40)$$

By the Cauchy–Schwarz inequality applied to the vectors $(\sigma_1, \dots, \sigma_d)$ and $(1, \dots, 1)$:

$$\|Z_T^\top Z_P\|_F = \sqrt{\sum_i \sigma_i^2} \leq \sum_i \sigma_i = \|Z_T^\top Z_P\|_*, \quad (41)$$

since $\|\mathbf{x}\|_1 \geq \|\mathbf{x}\|_2$ for any non-negative vector. Dividing both sides by $\|Z_T\|_F \|Z_P\|_F$ gives $\Omega_F \leq \Omega_N$.

Consequence for Risk Bounding. Ω_F does not arise from the alignment residual $\|Z_T - Z_P W\|_F^2$ at any $W \in \mathcal{O}(d)$ (which produces only trace-based similarities Ω_W); we treat it as an external comparison object. Substituting Ω_F into the bound formula in place of Ω_W yields

$$(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_F) \geq (\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_N), \quad (42)$$

i.e., a *strictly looser* value than the alignment-derived nuclear-form term. Among the alignment-derived members $\{\Omega_W\}$, the nuclear form Ω_N is the tightest because it exactly solves $\max_{W \in \mathcal{O}(d)} \text{Tr}(Z_T^\top Z_P W)$; the trace form Ω used in the main text (Eq. 4) satisfies $\Omega \leq \Omega_N$ in general, with equality on $Z_T^\top Z_P$ symmetric PSD (Appendix C.2).

Relation to CKA. Linear CKA [16] is defined as $\text{CKA}(Z_T, Z_P) = \|Z_T^\top Z_P\|_F^2 / (\|Z_T^\top Z_T\|_F \cdot \|Z_P^\top Z_P\|_F)$, which differs from $\Omega_F^2 = \|Z_T^\top Z_P\|_F^2 / (\|Z_T\|_F^2 \|Z_P\|_F^2)$ in its denominator. The two denominators are related by the inequality $\|Z_M^\top Z_M\|_F \leq \|Z_M\|_F^2 = \text{Tr}(Z_M^\top Z_M)$ (since $\|A\|_F \leq \text{Tr}(A)$ for any positive semidefinite A , by Cauchy–Schwarz on the eigenvalues), which gives $\text{CKA} \geq \Omega_F^2$. Hence CKA can overestimate alignment relative to Ω_F , and it is not directly substitutable into our Procrustes residual. More fundamentally, CKA is not derived from any alignment optimization and therefore lacks the alignment-grounded meaning of Ω_N , which exactly solves the Orthogonal Procrustes problem.

C Extension to General Orthogonal Alignments

The main text (Theorem 1) states the bound at the identity alignment $W = I$ (trace form Ω , Eq. 4). Here we present the general formulation where W is an arbitrary orthogonal matrix, recover both

the $W = I$ specialization and the Procrustes-optimal $W = W_N$ specialization (nuclear form Ω_N , used in the Sec. 5.5 ablation), and discuss the joint optimization over feature and head terms. The two specializations encode different priorities: W_N minimizes the feature alignment residual but generally inflates the head term, while $W = I$ keeps the head term at its minimum (vanishing exactly under frozen `lm_head`) and is autograd-compatible without SVD, properties that the LLM settings studied here—PTQ with FP16 head and frozen-head LoRA—require.

C.1 The General Bound for Arbitrary W

The exact geometric decomposition (Proposition 1) generalizes to any $W \in \mathcal{O}(d)$. Expanding the squared Frobenius norm and using the orthogonality of W :

$$\frac{1}{n} \|Z_T - Z_P W\|_F^2 = (\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W), \quad (43)$$

where $\Omega_W(Z_T, Z_P)$ is the *alignment-dependent similarity*:

$$\Omega_W(Z_T, Z_P) := \frac{\text{Tr}(Z_T^\top Z_P W)}{\|Z_T\|_F \|Z_P\|_F}. \quad (44)$$

This yields a family of risk bounds parameterized by W :

$$|\mathcal{R}_T - \mathcal{R}_P| \leq \underbrace{K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega_W)}}_{\delta(W)} + \underbrace{K_{\text{pred}} \|\Sigma_P^{1/2} (W H_T - H_P)\|_F}_{\gamma(W)}. \quad (45)$$

The Procrustes solution W_N minimizes $\delta(W)$ alone, yielding $\Omega_{W_N} = \Omega_N$ (the nuclear-norm similarity, Eq. 38). However, W_N enters the head term $\gamma(W_N)$, which can be inflated when the rotation is applied to a nearly preserved head ($H_T \approx H_P$). This creates a fundamental trade-off: W_N gives the tightest feature bound but may worsen the head bound, while $W = I$ preserves head coherence at the cost of a slightly looser feature bound.

C.2 The Trace Specialization ($W = I$)

Setting $W = I$, the alignment-dependent similarity reduces to the main-text *trace similarity* Ω (Eq. 4):

$$\Omega = \Omega_{W=I} = \frac{\text{Tr}(Z_T^\top Z_P)}{\|Z_T\|_F \|Z_P\|_F}. \quad (46)$$

Ordering of Similarities. Since the Procrustes solution maximizes $\text{Tr}(Z_T^\top Z_P W)$ over $\mathcal{O}(d)$:

$$\text{Tr}(Z_T^\top Z_P) \leq \|Z_T^\top Z_P\|_*, \quad (47)$$

which gives $\Omega \leq \Omega_N$. Equality holds if and only if $W_N = I$, i.e., $Z_T^\top Z_P$ is symmetric positive semidefinite.

When does $\Omega = \Omega_N$? Let $Z_T^\top Z_P = U \Sigma V^\top$ be the SVD. Then $\Omega = \Omega_N$ iff $V U^\top = I$, i.e., $U = V$, which holds when $Z_T^\top Z_P$ is symmetric. In post-training quantization with mild noise, $Z_F^\top Z_Q \approx Z_F^\top Z_F$ is approximately SPSPD, so $\Omega \approx \Omega_N$. As quantization becomes more aggressive, asymmetric perturbations cause Ω to fall below Ω_N , and this gap reflects genuine asymmetric distortion that a rotation-invariant metric would mask.

Full Bound under $W = I$. The identity alignment yields the main-text Theorem 1:

$$|\mathcal{R}_T - \mathcal{R}_P| \leq K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T \rho_P (1 - \Omega)} + K_{\text{pred}} \|\Sigma_P^{1/2} (H_T - H_P)\|_F. \quad (48)$$

The head term $\|H_T - H_P\|$ carries no rotational contamination and vanishes whenever the prediction head is preserved (FP16 head under PTQ, frozen head under LoRA).

Relation to CKA. Both the trace form Ω and the Frobenius similarity Ω_F (Appendix B) are bounded above by the nuclear Ω_N (since $\text{Tr}(M) \leq \|M\|_*$ and $\|M\|_F \leq \|M\|_*$), but Ω and Ω_F do not have a fixed ordering relative to each other. Linear CKA avoids explicit rotational alignment like Ω , but uses the Frobenius norm rather than the trace, and lacks a direct interpretation as an alignment residual.

C.3 Joint Optimization on the Stiefel Manifold

The tightest bound over all orthogonal alignments requires solving:

$$W_{\text{opt}} = \arg \min_{W \in \mathcal{O}(d)} \left[\underbrace{K_{\text{feat}} \sqrt{\frac{1}{n} \|Z_T - Z_P W\|_F^2}}_{\delta(W)} + \underbrace{K_{\text{pred}} \|\Sigma_P^{1/2} (W H_T - H_P)\|_F}_{\gamma(W)} \right]. \quad (49)$$

Unlike the standard Procrustes problem (which minimizes δ alone and admits the closed-form $W_N = V U^\top$), the addition of $\gamma(W)$ introduces a quadratic dependence on W through the head term, related to the *Weighted Orthogonal Procrustes Problem* (WOPP) [7].

Why No Closed-Form Solution Exists. The feature term $\delta(W)$ is concave in $\text{Tr}(Z_T^\top Z_P W)$, while $\gamma(W)$ is convex in W . Their sum is neither convex nor concave on $\mathcal{O}(d)$, precluding a simple variational characterization.

Special Case: Isotropic Features. If $\Sigma_P \approx \lambda I$, the head term becomes independent of W and the joint objective reduces to a standard Procrustes problem. However, the isotropic assumption is unrealistic for LLM representations, which exhibit well-documented anisotropy [5, 3].

Numerical Solution. For the general case, W_{opt} can be approximated via Riemannian gradient descent on the Stiefel manifold, but this entangles scale, shape, and head components, destroying the clean physical interpretation.

Practical Irrelevance for This Paper’s Regimes. In the two primary settings:

- **Quantization** ($H_T \approx H_P$): The head term is negligible, so $W_{\text{opt}} \approx W_N$ and the Procrustes-optimal solution suffices.
- **LoRA** ($H_T = H_P$): The head term vanishes for any W , so the joint problem reduces to standard Procrustes.

The Riemannian formulation becomes relevant only when comparing models with *both* rotated features and divergent heads (e.g., full-parameter SFT). We leave this setting to future work.

D Extension to Autoregressive Generation

The bound of Theorem 1 applies to any matrix pair $(Z_T, Z_P) \in \mathbb{R}^{n \times d}$ of feature vectors, whatever the origin of the rows. We spell out here the autoregressive instantiation informally summarized in Sec. 3.4.

From tokens to matrices. Under teacher forcing, each target token y_τ in a sequence (c, y) contributes an independent feature–loss pair $(\phi_M(c, y_{<\tau}), \ell(\phi_M(c, y_{<\tau}) H_M, y_\tau))$. The per-sequence average inside Eq. (7) is therefore the empirical mean of these pairs over the $|y|$ positions, and the expectation over sequences becomes an empirical mean over all $N = \sum_{(c,y)} |y|$ tokens in the calibration set. Stacking the token-level features into a matrix $Z_M^{\text{AR}} \in \mathbb{R}^{N \times d}$ therefore reduces the autoregressive risk gap to exactly the per-row setting that Theorem 1 controls.

Corollary 1 (Autoregressive Risk Bound). *Let $Z_T^{\text{AR}}, Z_P^{\text{AR}} \in \mathbb{R}^{N \times d}$ be the token-level feature matrices collected under teacher forcing, with RMS scales $\rho_T^{\text{AR}}, \rho_P^{\text{AR}}$, Procrustes similarity Ω^{AR} , and covariance Σ_P^{AR} . Then*

$$|\mathcal{R}_T^{\text{AR}} - \mathcal{R}_P^{\text{AR}}| \leq K_{\text{feat}} \sqrt{(\rho_T^{\text{AR}} - \rho_P^{\text{AR}})^2 + 2 \rho_T^{\text{AR}} \rho_P^{\text{AR}} (1 - \Omega^{\text{AR}})} + K_{\text{pred}} \|(\Sigma_P^{\text{AR}})^{1/2} (H_T - H_P)\|_F. \quad (50)$$

Proof. Each row of Z_M^{AR} is a feature vector of dimension d , and the Lipschitz analysis of Appendix A.3 together with the Procrustes decomposition of Appendix A.4 depend only on the matrix shape, not on the provenance of its rows. Applying Theorem 1 to $(Z_T^{\text{AR}}, Z_P^{\text{AR}})$ yields the stated inequality. $\square$

This unification enables one bound to cover point-wise classification (MMLU, ARC, where $|y|=1$), language modeling (WikiText, FineWeb-Edu, where y is a long chunk), short-horizon QA (SQuAD, TriviaQA, where y is a 1–5 token span), and multi-step reasoning (GSM8K, where y is a chain-of-thought solution).

E Model and Quantization Details

E.1 Target Models

Table 4 lists all target (full-precision) models evaluated in the quantization experiments. All targets are loaded in BF16 as the reference precision; an FP16 proxy is additionally included for each model to measure the BF16→FP16 precision-drift baseline.

Table 4: Target models used in quantization experiments. All models are loaded in BF16.

#	Family	HuggingFace ID	Variant	Params	d	$ V $
1	Qwen 3	Qwen/Qwen3-8B-Base	Base	8B	4096	151K
2	Llama 3.1	meta-llama/Meta-Llama-3.1-8B	Base	8B	4096	128K
3	Ministral 3	mistralai/Ministral-3-8B-Base-2512	Base	8B	4096	131K
4	Qwen 3	Qwen/Qwen3-8B	Instruct	8B	4096	151K
5	Llama 3.1	meta-llama/Meta-Llama-3.1-8B-Instruct	Instruct	8B	4096	128K
6	Ministral 3	mistralai/Ministral-3-8B-Instruct-2512	Instruct	8B	4096	131K
7	DeepSeek R1	deepseek-ai/DeepSeek-R1-Distill-Llama-8B	Distilled	8B	4096	128K

Models 2, 5 (meta-llama) are gated and require accepting the Meta Community License on HuggingFace before access.

E.2 Quantization Backends and Bit-Widths

Every target model is evaluated against up to four quantization backends. Table 5 summarises the shared GGUF and BitsAndBytes (BnB) configurations applied uniformly across all models.

Table 5: Shared quantization tiers applied to all target models. GGUF quants use the llama.cpp k-quant scheme; BnB quants use on-the-fly quantization from the full-precision checkpoint.

Tier	Backend	Tag	Description
FP16 reference	dtype	dtype:float16	BF16→FP16 precision drift baseline
8-bit	GGUF	Q8_0	Round-to-nearest, 8-bit
	BnB	bnb:int8	LLM.int8() absmax
6-bit	GGUF	Q6_K	k-quant, 6-bit (super-blocks)
5-bit	GGUF	Q5_K_M	k-quant, 5-bit (medium)
4-bit	GGUF	Q4_K_M	k-quant, 4-bit (medium)
	BnB	bnb:nf4	NormalFloat4 (QLoRA format)
	BnB	bnb:fp4	FP4 with FP16 compute
3-bit	GGUF	Q3_K_M	k-quant, 3-bit (medium)
2-bit	GGUF	Q2_K	k-quant, 2-bit

E.3 Per-Model GPTQ Repositories

Table 6 lists the pre-quantized GPTQ checkpoint used for each target model. All repositories are publicly hosted on HuggingFace and loaded via the standard transformers API unless otherwise noted.

E.4 GGUF Repositories

Table 7 lists the GGUF repositories and filename conventions for each model. GGUF files follow one of two naming conventions depending on the repository provider: *dot conven-*

Table 6: Pre-quantized GPTQ checkpoints per model. One repository per model, corresponding to the variant actually loaded in our experiments; “—” indicates no GPTQ checkpoint was evaluated for that model. †Post-quantization fine-tuned on distillation data; may not reflect clean quantization error.

#	Model	GPTQ Repository
1	Qwen3-8B-Base	AlphaGa0/Qwen3-8B-GPTQ [†]
2	Llama-3.1-8B	ModelCloud/Meta-Llama-3.1-8B-gptq-4bit
3	Ministral-3-Base	—
4	Qwen3-8B	JunHowie/Qwen3-8B-GPTQ-Int8
5	Llama-3.1-8B-Inst.	ModelCloud/Meta-Llama-3.1-8B-Instruct-gptq-4bit
6	Ministral-3-Inst.	—
7	DeepSeek-R1-Distill	jakiAJK/DeepSeek-R1-Distill-Llama-8B-GPTQ-int4
8	Qwen2.5-7B	—
9	Qwen2.5-7B-Inst.	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4

695 *tion* (Model.Q4_K_M.gguf) used by QuantFactory and mradermacher, or *dash convention*
696 (Model-Q4_K_M.gguf) used by Qwen, bartowski, and most official releases.

Table 7: GGUF repositories and filename templates. Each repository provides quant8_0, Q6_K, Q5_K_M, Q4_K_M, Q3_K_M, and Q2_K unless otherwise noted.

#	Model	GGUF Repository	Filename Template
1	Qwen3-8B-Base	mradermacher/Qwen3-8B-Base-GGUF	Qwen3-8B-Base.{Q}.gguf
2	Llama-3.1-8B	QuantFactory/Meta-Llama-3.1-8B-GGUF	Meta-Llama-3.1-8B.{Q}.gguf
3	Ministral-3-Base	mradermacher/Ministral-3-8B-Base-2512-GGUF	Ministral-3-8B-Base-2512.{Q}.gguf
4	Qwen3-8B	Qwen/Qwen3-8B-GGUF	Qwen3-8B-{Q}.gguf
5	Llama-3.1-8B-Inst.	bartowski/Meta-Llama-3.1-8B-Instruct-GGUF	...-Instruct-{Q}.gguf
6	Ministral-3-Inst.	bartowski/mistralai_Ministral-3-8B-Instruct-2512-GGUF	...-2512-{Q}.gguf
7	DeepSeek-R1-Distill	bartowski/DeepSeek-R1-Distill-Llama-8B-GGUF	...-Llama-8B-{Q}.gguf
8	Qwen2.5-7B	QuantFactory/Qwen2.5-7B-GGUF	Qwen2.5-7B.{Q}.gguf
9	Qwen2.5-7B-Inst.	Qwen/Qwen2.5-7B-Instruct-GGUF	...-Instruct-{Q}.gguf

697 E.5 Backend Coverage Summary

698 Table 8 provides a per-model summary of quantization backend coverage across the combinations we
699 evaluated.

Table 8: Quantization backend coverage per model. ✓ = included in our experiments; — = not evaluated in this study.

#	Model	BnB	GGUF	GPTQ
1	Qwen3-8B-Base	✓	✓	✓
2	Llama-3.1-8B	✓	✓	✓
3	Ministral-3-8B-Base	✓	✓	—
4	Qwen3-8B	✓	✓	✓
5	Llama-3.1-8B-Instruct	✓	✓	✓
6	Ministral-3-8B-Instruct	✓	✓	—
7	DeepSeek-R1-Distill-Llama	✓	✓	✓
8	Qwen2.5-7B	✓	✓	—
9	Qwen2.5-7B-Instruct	✓	✓	✓
Total		9/9	9/9	6/9

700 **Experiment scope.** Each model is evaluated on the backend/bit-width combinations marked ✓ in
701 Table 8: GGUF (6 bit-widths) and BitsAndBytes (3 configurations) on every model, plus the GPTQ
702 checkpoint listed in Table 6 where available. The cells marked “—” are outside our experiment scope
703 under the compute and time budget for this study.

F Quantization Replication, Feature-Only Scatter, and Per-Model Tables

F.1 Replication on Ministral and DeepSeek

Figure 5 replicates the main quantization scatter (Fig. 2) on the two additional 8B families—Ministral-3-8B and DeepSeek-R1-Distill-Llama-8B—kept out of the main text for symmetry with the LoRA experiments (Sec. 5.1). The bound tracks $|\Delta\mathcal{R}|$ on these two families with patterns consistent with the Llama–Qwen main analysis, supporting the cross-family generality of the PRISM framework. Per-variant numerical decompositions for all four families are in Appendix F.3.

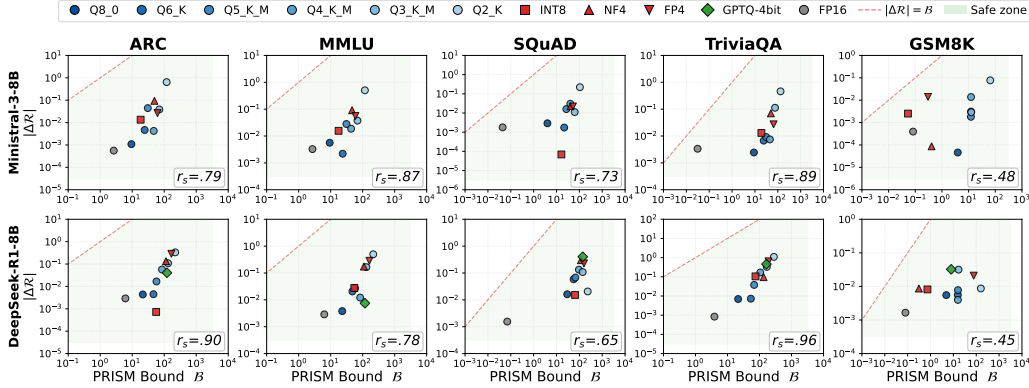


Figure 5: **PTQ replication on Ministral-3-8B and DeepSeek-R1-Distill-Llama-8B.** Counterpart of main-text Fig. 2 (Llama-3.1-8B and Qwen3-8B), with identical axes, PTQ family color coding, and benchmark set. Rows: Ministral-3-8B, DeepSeek-R1-Distill-Llama-8B. Columns: ARC, MMLU, SQuAD, TriviaQA, GSM8K. Per-subplot Spearman r_s is annotated in each panel.

F.2 Feature-Only Scatter

Figure 6 reports the feature-alignment-only replication of the main quantization result (Fig. 2), with the x-axis replaced by the head-free term $\delta = K_{\text{feat}} \sqrt{(\rho_T - \rho_P)^2 + 2\rho_T\rho_P(1 - \Omega)}$. The grid isolates the backbone contribution to the bound and is the source of the aggregate feature-only correlation $\overline{r_s}(\delta)$ cited in Sec. 5.2.

F.3 Per-Model Decomposition Tables

We report the full geometric decomposition (Eq. 3) for each evaluated model, covering all quantization variants and benchmarks. Each table column lists the base RMS scale ρ_T , proxy RMS scale ρ_P , identity-alignment similarity Ω , feature alignment error δ , head discrepancy γ , full PRISM bound B , and empirical cross-entropy gap $|\Delta\mathcal{R}|$. Per-dataset Spearman rank correlations $r_s(\delta, |\Delta\mathcal{R}|)$ appear in the left-hand column of each benchmark block.

The base-model tables cover the four 8B families: Llama-3.1-8B extended benchmarks (WikiText, FineWeb-Edu) in Table 9, Ministral-3-8B in Table 10, Qwen3-8B in Table 11 (also referenced in Sec. 5.3), and DeepSeek-R1-Distill-Llama-8B in Table 12. The instruction-tuned counterparts of Llama, Ministral, and Qwen are reported in Tables 13, 14, and 15 respectively, and reproduce the same patterns as their base counterparts.

Per-benchmark variability in correlation strength. Among MC/QA/reasoning benchmarks the bound consistently achieves $|r_s| \in [0.79, 0.95]$ across all evaluated families. Language-modeling benchmarks show more variability: FineWeb-Edu retains $|r_s| \approx 0.92$ while WikiText is weaker ($|r_s| \approx 0.54$). The contrast suggests the WikiText weakness is dataset-specific—likely the dense per-token averaging interacting with WikiText’s specific text distribution—rather than a structural limitation of LM evaluation, since FineWeb-Edu (also LM scoring) tracks the bound as well as the MC/QA benchmarks.

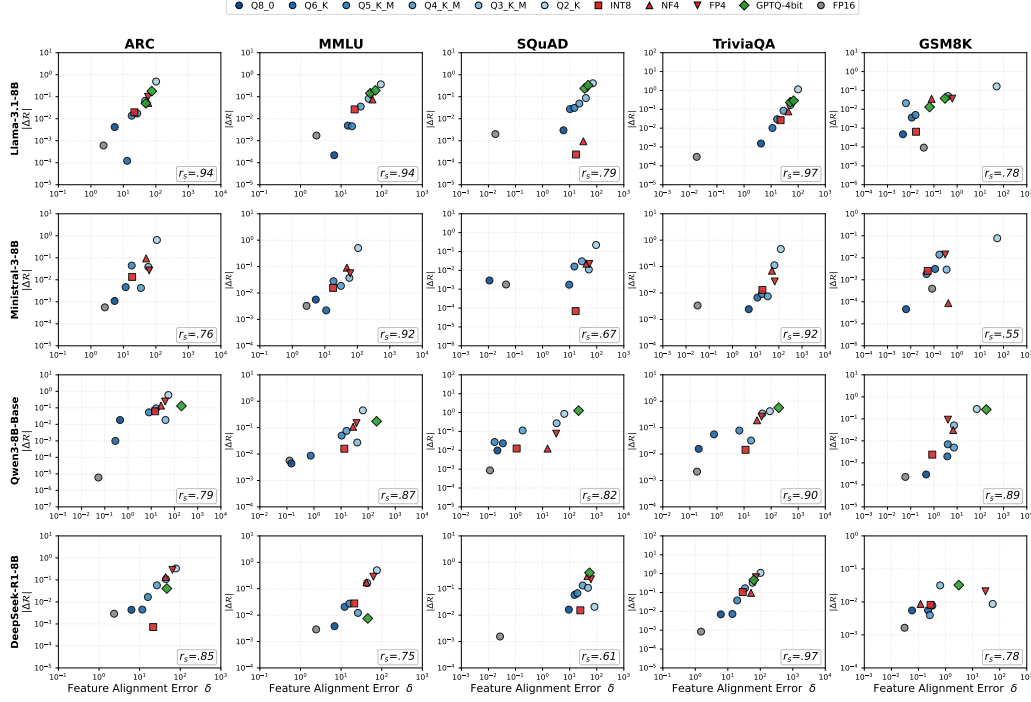


Figure 6: **Feature alignment error δ alone is already highly predictive of the risk gap.** Identical layout to Fig. 2 but with x-axis replaced by the feature-only term δ . Because most protocols retain the `lm_head` in FP16, the head-discrepancy term γ is zero for GPTQ and BitsAndBytes and small for GGUF configurations that quantize output embeddings; δ therefore captures most of the predictive signal of Fig. 2, confirming that scale collapse $(\Delta\rho)^2$ and shape mismatch $(1 - \Omega)$ are the dominant degradation channels under PTQ (Sec. 4).

Table 9: Geometric decomposition for **Llama-3.1-8B** under identity alignment ($W=I$) — ext task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.92$)	GGUF	FP16	137.58	137.63	0.9998	2.4488	0	6.4005	0.0006
		Q8_0	137.58	137.68	0.9992	5.4060	3.8397	19.5600	0.0042
		Q6_K	137.58	138.13	0.9955	13.0869	12.0511	51.2485	0.0001
		Q5_K_M	137.58	138.57	0.9915	18.0240	12.0975	64.2184	0.0137
		Q4_K_M	137.58	137.71	0.9812	26.6728	12.0257	86.7227	0.0173
		Q3_K_M	137.58	138.68	0.9420	47.0681	12.0855	140.1155	0.0654
	BnB	Q2_K	137.58	144.67	0.7408	101.8302	12.7089	284.1309	0.4877
		INT8	137.58	137.99	0.9874	21.8568	0	57.1279	0.0195
		NF4	137.58	142.78	0.9155	57.8645	0	151.2430	0.0523
		FP4	137.58	135.87	0.9117	57.4939	0	150.2742	0.0992
	GPTQ	GPTQ-4bit	137.58	138.70	0.9363	49.3164	0	128.9004	0.0501
SQuAD ($r_s=0.73$)	GGUF	FP16	138.24	138.22	1.0000	0.0181	0	0.0474	0.0020
		Q8_0	138.24	138.17	0.9991	5.9183	3.7829	20.8187	0.0030
		Q6_K	138.24	138.21	0.9972	10.3184	11.9097	43.8126	0.0274
		Q5_K_M	138.24	138.17	0.9941	14.9613	11.9042	55.9400	0.0310
		Q4_K_M	138.24	138.17	0.9865	22.7357	11.9062	76.2632	0.0488
		Q3_K_M	138.24	138.24	0.9589	39.6157	11.9122	120.3916	0.0870
	BnB	Q2_K	138.24	138.53	0.8655	71.7843	11.9300	204.4973	0.4090
		INT8	138.24	138.14	0.9924	17.0726	0	44.6233	0.0002
		NF4	138.24	138.33	0.9731	32.0471	0	83.7629	0.0009
		FP4	138.24	139.19	0.9466	45.3601	0	118.5596	0.2508

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
	GPTQ	GPTQ-4bit	138.24	138.33	0.9663	35.9111	0	93.8623	0.2382
GSM8K ($r_s=0.73$)	–	FP16	134.24	134.28	1.0000	0.0366	0	0.0956	0.0001
		GGUF Q8_0	134.24	134.24	1.0000	0.0046	3.8092	5.3992	0.0005
		Q6_K	134.24	134.23	1.0000	0.0111	11.9756	16.9652	0.0036
		Q5_K_M	134.24	134.26	1.0000	0.0160	11.9769	16.9797	0.0050
		Q4_K_M	134.24	134.24	1.0000	0.0062	11.9732	16.9488	0.0211
		Q3_K_M	134.24	134.65	1.0000	0.4103	12.0185	18.0690	0.0502
	BnB	Q2_K	134.24	134.51	0.9298	50.3457	11.9866	148.5423	0.1630
		INT8	134.24	134.22	1.0000	0.0169	0	0.0441	0.0006
		NF4	134.24	134.32	1.0000	0.0794	0	0.2076	0.0356
		FP4	134.24	134.86	1.0000	0.6186	0	1.6168	0.0367
	GPTQ	GPTQ-4bit	134.24	134.18	1.0000	0.0648	0	0.1695	0.0132
WikiText ($r_s=0.65$)	–	FP16	138.98	139.00	1.0000	0.0287	0	0.0751	0.0017
		GGUF Q8_0	138.98	138.96	1.0000	0.0171	3.9304	5.6032	0.0015
		Q6_K	138.98	139.02	1.0000	0.0421	12.2433	17.4247	0.0079
		Q5_K_M	138.98	139.00	1.0000	0.0228	12.2409	17.3710	0.0081
		Q4_K_M	138.98	138.99	1.0000	0.0179	12.2409	17.3581	0.0011
		Q3_K_M	138.98	139.22	0.9919	17.7597	12.2705	63.7722	0.0029
	BnB	Q2_K	138.98	138.96	0.8690	71.1281	12.2364	203.2154	0.3367
		INT8	138.98	138.96	1.0000	0.0183	0	0.0478	0.0086
		NF4	138.98	138.88	1.0000	0.0922	0	0.2411	0.0491
		FP4	138.98	139.44	0.9785	28.8896	0	75.5100	0.1051
	GPTQ	GPTQ-4bit	138.98	138.96	0.9999	2.3927	0	6.2540	0.0037
FineWeb-Edu ($r_s=0.53$)	–	FP16	109.80	109.85	1.0000	0.0575	0	0.1502	0.0003
		GGUF Q8_0	109.80	109.79	1.0000	0.0069	3.9676	5.6290	0.0019
		Q6_K	109.80	109.79	1.0000	0.0051	12.3239	17.4419	0.0040
		Q5_K_M	109.80	109.80	1.0000	0.0057	12.3255	17.4459	0.0087
		Q4_K_M	109.80	109.74	1.0000	0.0551	12.3085	17.5508	0.0259
		Q3_K_M	109.80	109.75	1.0000	0.0430	12.3148	17.5281	0.0860
	BnB	Q2_K	109.80	109.55	1.0000	0.2479	12.2645	17.9927	0.3633
		INT8	109.80	109.76	1.0000	0.0323	0	0.0844	0.0112
		NF4	109.80	109.72	1.0000	0.0805	0	0.2103	0.0551
		FP4	109.80	109.75	1.0000	0.0421	0	0.1099	0.1034
	GPTQ	GPTQ-4bit	109.80	109.73	1.0000	0.0616	0	0.1610	0.0600

Table 10: Geometric decomposition for **Ministral-3-8B-Base** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.76$)	–	FP16	191.14	191.11	0.9999	2.6886	0	2.6480	0.0006
		GGUF Q8_0	191.14	191.06	0.9996	5.3670	2.9300	9.4295	0.0011
		Q6_K	191.14	191.08	0.9981	11.7814	9.1076	24.4836	0.0047
		Q5_K_M	191.14	191.55	0.9955	18.0613	9.1239	30.6917	0.0443
		Q4_K_M	191.14	190.56	0.9835	34.7202	9.0663	47.0176	0.0042
		Q3_K_M	191.14	189.89	0.9529	58.4704	9.0443	70.3780	0.0387
	BnB	Q2_K	191.14	194.14	0.8439	107.6601	9.1676	118.9992	0.6424
		INT8	191.14	191.46	0.9953	18.6049	0	18.3240	0.0135
		NF4	191.14	190.54	0.9655	50.1598	0	49.4023	0.0936
		FP4	191.14	191.92	0.9468	62.4786	0	61.5350	0.0265
MMLU ($r_s=0.92$)	–	GGUF FP16	191.06	191.04	0.9999	2.7396	0	2.6983	0.0033
		Q8_0	191.06	191.04	0.9996	5.2595	2.9475	9.3485	0.0056
		Q6_K	191.06	191.13	0.9983	11.0202	9.1283	23.7632	0.0022
		Q5_K_M	191.06	191.55	0.9954	18.4222	9.1356	31.0637	0.0280
		Q4_K_M	191.06	191.12	0.9867	31.2214	9.1089	43.6318	0.0187

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
	BnB	Q3_K_M	191.06	190.43	0.9555	56.8884	9.0784	68.8680	0.0374
		Q2_K	191.06	193.99	0.8483	106.0716	9.2112	117.4963	0.5035
		INT8	191.06	191.21	0.9956	17.9674	0	17.6961	0.0156
		NF4	191.06	190.92	0.9691	47.4992	0	46.7819	0.0910
		FP4	191.06	191.57	0.9503	60.3029	0	59.3922	0.0558
SQuAD ($r_s=0.67$)	GGUF	FP16	186.67	186.62	1.0000	0.0447	0	0.0440	0.0018
		Q8_0	186.67	186.68	1.0000	0.0109	2.8244	4.0050	0.0029
		Q6_K	186.67	186.66	0.9987	9.6602	8.6258	21.7130	0.0017
		Q5_K_M	186.67	186.60	0.9968	14.9459	8.6226	26.9145	0.0160
		Q4_K_M	186.67	186.82	0.9885	28.3714	8.6230	40.1377	0.0302
	BnB	Q3_K_M	186.67	186.89	0.9626	51.0640	8.6352	62.5048	0.0111
		Q2_K	186.67	188.53	0.8724	94.7993	8.6737	105.6341	0.2206
		INT8	186.67	186.71	0.9960	16.5948	0	16.3442	0.0001
		NF4	186.67	186.50	0.9748	41.8712	0	41.2389	0.0226
		FP4	186.67	187.23	0.9598	53.0339	0	52.2330	0.0219
TriviaQA ($r_s=0.92$)	GGUF	FP16	194.05	194.02	1.0000	0.0315	0	0.0311	0.0034
		Q8_0	194.05	194.07	0.9997	5.0362	2.8683	9.0165	0.0024
		Q6_K	194.05	194.09	0.9981	11.8858	8.7989	24.1498	0.0067
		Q5_K_M	194.05	194.13	0.9954	18.6732	8.7969	30.8319	0.0092
		Q4_K_M	194.05	194.23	0.9854	33.1589	8.8004	45.1038	0.0075
	BnB	Q3_K_M	194.05	193.98	0.9452	64.2222	8.7925	75.6868	0.1125
		Q2_K	194.05	195.16	0.8057	121.3231	8.8470	132.0025	0.4595
		INT8	194.05	194.15	0.9950	19.4413	0	19.1477	0.0130
		NF4	194.05	194.10	0.9661	50.5577	0	49.7942	0.0697
		FP4	194.05	194.43	0.9417	66.3057	0	65.3043	0.0272
GSM8K ($r_s=0.55$)	GGUF	FP16	179.04	179.13	1.0000	0.0838	0	0.0826	0.0004
		Q8_0	179.04	179.05	1.0000	0.0063	2.7631	3.9138	0.0000
		Q6_K	179.04	178.93	1.0000	0.1124	8.6061	12.2816	0.0031
		Q5_K_M	179.04	179.00	1.0000	0.0477	8.6085	12.2213	0.0018
		Q4_K_M	179.04	179.22	1.0000	0.1722	8.6125	12.3495	0.0139
	BnB	Q3_K_M	179.04	179.40	1.0000	0.3526	8.6253	12.5452	0.0029
		Q2_K	179.04	181.13	0.9561	53.3729	8.6684	64.8258	0.0772
		INT8	179.04	179.10	1.0000	0.0539	0	0.0531	0.0025
		NF4	179.04	178.63	1.0000	0.4159	0	0.4096	0.0001
		FP4	179.04	179.35	1.0000	0.3057	0	0.3011	0.0141
WikiText ($r_s=0.84$)	GGUF	FP16	186.63	186.68	1.0000	0.0495	0	0.0488	0.0002
		Q8_0	186.63	186.65	1.0000	0.0187	2.8036	3.9833	0.0008
		Q6_K	186.63	186.65	1.0000	0.0227	8.6688	12.2819	0.0090
		Q5_K_M	186.63	186.66	1.0000	0.0313	8.6676	12.2886	0.0068
		Q4_K_M	186.63	186.77	1.0000	0.1366	8.6676	12.3924	0.0113
	BnB	Q3_K_M	186.63	186.76	1.0000	0.1351	8.6715	12.3964	0.0549
		Q2_K	186.63	187.31	0.9050	81.5046	8.6981	92.5746	0.2368
		INT8	186.63	186.71	1.0000	0.0844	0	0.0831	0.0229
		NF4	186.63	186.53	1.0000	0.0955	0	0.0940	0.0475
		FP4	186.63	186.84	1.0000	0.2104	0	0.2072	0.0635
FineWeb-Edu ($r_s=0.49$)	GGUF	FP16	147.77	147.98	1.0000	0.2143	0	0.2111	0.0003
		Q8_0	147.77	147.78	1.0000	0.0057	2.7993	3.9644	0.0001
		Q6_K	147.77	147.77	1.0000	0.0011	8.6771	12.2723	0.0014
		Q5_K_M	147.77	147.75	1.0000	0.0181	8.6754	12.2867	0.0034
		Q4_K_M	147.77	147.80	1.0000	0.0334	8.6772	12.3043	0.0134
	BnB	Q3_K_M	147.77	147.66	1.0000	0.1058	8.6741	12.3713	0.0510
		Q2_K	147.77	147.59	1.0000	0.1815	8.6909	12.4696	0.2140
		INT8	147.77	147.78	1.0000	0.0074	0	0.0073	0.0030
		NF4	147.77	147.62	1.0000	0.1542	0	0.1519	0.0288
		FP4	147.77	147.69	1.0000	0.0834	0	0.0822	0.0513

Table 11: Geometric decomposition for **Qwen3-8B-Base** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.79$)	GGUF	FP16	332.71	332.77	1.0000	0.0529	0	0.1831	0.0000
		Q8_0	332.71	332.26	1.0000	0.4528	18.9175	28.3194	0.0183
		Q6_K	332.71	332.43	1.0000	0.2856	60.1717	86.0832	0.0010
		Q5_K_M	332.71	333.21	0.9997	7.9547	60.2790	112.7570	0.0533
		Q4_K_M	332.71	333.72	0.9988	16.2022	60.3711	141.4092	0.0932
		Q3_K_M	332.71	319.87	0.9927	41.3976	57.6558	224.7021	0.0184
	BnB	Q2_K	332.71	338.59	0.9869	54.6969	61.2316	275.7516	0.5978
		INT8	332.71	329.14	0.9991	14.7291	0	50.9372	0.0620
		NF4	332.71	328.25	0.9969	26.3152	0	91.0051	0.1367
		FP4	332.71	333.50	0.9929	39.7238	0	137.3760	0.2419
MMLU ($r_s=0.87$)	GGUF	FP16	332.87	333.00	1.0000	0.1246	0	0.4310	0.0057
		Q8_0	332.87	332.73	1.0000	0.1463	18.9548	27.3120	0.0044
		Q6_K	332.87	332.12	1.0000	0.7540	60.1581	87.6841	0.0087
		Q5_K_M	332.87	332.02	0.9995	10.4878	60.1161	121.2869	0.0500
		Q4_K_M	332.87	333.30	0.9989	15.8100	60.3727	140.0554	0.0754
		Q3_K_M	332.87	324.42	0.9931	39.5282	58.6394	219.6281	0.0274
	BnB	Q2_K	332.87	336.88	0.9819	63.7574	60.9749	306.7224	0.4521
		INT8	332.87	332.24	0.9992	13.1825	0	45.5886	0.0159
		NF4	332.87	329.71	0.9966	27.6565	0	95.6438	0.1076
		FP4	332.87	332.87	0.9938	37.1370	0	128.4300	0.1483
SQuAD ($r_s=0.82$)	GGUF	FP16	298.04	297.92	1.0000	0.1143	0	0.3954	0.0009
		Q8_0	298.04	298.25	1.0000	0.2145	16.9415	24.7008	0.0098
		Q6_K	298.04	298.38	1.0000	0.3412	53.5754	76.9469	0.0234
		Q5_K_M	298.04	297.87	1.0000	0.1689	53.4805	76.2168	0.0275
		Q4_K_M	298.04	299.88	1.0000	1.8365	53.8708	82.5360	0.1121
		Q3_K_M	298.04	297.59	0.9938	33.0481	53.3522	189.7407	0.2722
	BnB	Q2_K	298.04	307.56	0.9786	63.3464	55.3594	297.3595	0.8582
		INT8	298.04	296.94	1.0000	1.1027	0	3.8134	0.0124
		NF4	298.04	292.80	0.9988	15.3047	0	52.9278	0.0123
		FP4	298.04	304.82	0.9947	31.8703	0	110.2163	0.0758
TriviaQA ($r_s=0.90$)	GGUF	FP16	241.01	240.83	1.0000	0.1833	0	0.6338	0.0022
		Q8_0	241.01	241.22	1.0000	0.2113	13.4137	19.7007	0.0158
		Q6_K	241.01	241.79	1.0000	0.7774	42.5021	62.7955	0.0556
		Q5_K_M	241.01	239.98	0.9996	6.6875	42.1234	82.6986	0.0776
		Q4_K_M	241.01	241.74	0.9972	18.1291	42.4684	122.7549	0.0325
		Q3_K_M	241.01	235.11	0.9804	47.5370	41.0700	222.4779	0.3429
	BnB	Q2_K	241.01	253.07	0.9351	89.8118	44.8357	374.0012	0.4199
		INT8	241.01	239.25	0.9989	11.3354	0	39.2009	0.0145
		NF4	241.01	241.11	0.9921	30.2526	0	104.6219	0.1900
		FP4	241.01	241.03	0.9832	44.1714	0	152.7570	0.2615
GSM8K ($r_s=0.89$)	GGUF	FP16	267.17	267.23	1.0000	0.0586	0	0.2028	0.0002
		Q8_0	267.17	266.71	1.0000	0.4610	15.3184	23.2578	0.0003
		Q6_K	267.17	270.95	1.0000	3.7862	49.0890	82.5161	0.0020
		Q5_K_M	267.17	271.11	1.0000	3.9450	49.1205	83.1096	0.0070
		Q4_K_M	267.17	274.46	1.0000	7.2943	49.7265	95.5495	0.0049
		Q3_K_M	267.17	274.49	1.0000	7.3208	49.7003	95.6042	0.0504
	BnB	Q2_K	267.17	313.41	0.9835	70.0507	56.9784	322.8344	0.2799
		INT8	267.17	268.01	1.0000	0.8477	0	2.9317	0.0024
		NF4	267.17	273.95	1.0000	6.7810	0	23.4507	0.0309
		FP4	267.17	271.17	1.0000	4.0054	0	13.8517	0.0915
WikiText ($r_s=0.54$)	GGUF	FP16	295.29	295.05	1.0000	0.2454	0	0.8487	0.0062
		Q8_0	295.29	295.07	1.0000	0.2226	16.9085	24.6823	0.0136
		Q6_K	295.29	295.71	1.0000	0.4170	53.5197	77.1304	0.0936

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
	BnB	Q5_K_M	295.29	294.96	1.0000	0.3333	53.3760	76.6379	0.0419
		Q4_K_M	295.29	293.82	1.0000	1.4707	53.1654	80.2733	0.1189
		Q3_K_M	295.29	291.95	1.0000	3.3439	52.7849	86.2134	0.3420
		Q2_K	295.29	312.94	0.9918	42.7574	56.8034	228.1993	0.9686
		INT8	295.29	295.98	1.0000	0.6875	0	2.3774	0.0279
		NF4	295.29	292.17	1.0000	3.1177	0	10.7818	0.2456
		FP4	295.29	295.34	1.0000	0.0494	0	0.1709	0.6045
FineWeb-Edu ($r_s=0.92$)	GGUF	FP16	296.53	296.39	1.0000	0.1404	0	0.4857	0.0014
		Q8_0	296.53	296.48	1.0000	0.0508	17.3136	24.6607	0.0011
		Q6_K	296.53	296.81	1.0000	0.2782	54.6083	78.1900	0.0097
		Q5_K_M	296.53	296.41	1.0000	0.1152	54.5390	77.5283	0.0136
		Q4_K_M	296.53	295.72	1.0000	0.8065	54.4148	79.7430	0.0334
	BnB	Q3_K_M	296.53	292.88	1.0000	3.6487	53.8957	88.8383	0.1308
		Q2_K	296.53	307.07	1.0000	10.5437	56.5028	116.3702	0.6126
		INT8	296.53	296.64	1.0000	0.1157	0	0.4001	0.0058
		NF4	296.53	295.11	1.0000	1.4154	0	4.8947	0.0747
		FP4	296.53	298.25	1.0000	1.7161	0	5.9349	0.1809

Table 12: Geometric decomposition for **DeepSeek-R1-Distill-Llama-8B** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.85$)	GGUF	FP16	129.06	129.02	0.9998	2.3568	0	6.1255	0.0029
		Q8_0	129.06	128.80	0.9988	6.2897	3.5672	21.3925	0.0044
		Q6_K	129.06	129.14	0.9959	11.6577	11.1787	46.1087	0.0045
		Q5_K_M	129.06	128.92	0.9924	15.9382	11.1661	57.2161	0.0167
		Q4_K_M	129.06	130.23	0.9789	26.6440	11.3039	85.2364	0.0569
		Q3_K_M	129.06	133.12	0.9431	44.3871	11.5452	131.6935	0.1086
	BnB	Q2_K	129.06	134.02	0.8228	78.4551	11.5442	220.2377	0.3327
		INT8	129.06	129.36	0.9861	21.5708	0	56.0644	0.0007
		NF4	129.06	132.28	0.9445	43.6391	0	113.4221	0.1337
		FP4	129.06	135.34	0.8825	64.3843	0	167.3408	0.2866
	GPTQ	GPTQ-4bit	129.06	132.05	0.9360	46.7936	0	121.6208	0.0407
MMLU ($r_s=0.75$)	GGUF	FP16	130.42	130.30	0.9998	2.4328	0	6.3231	0.0029
		Q8_0	130.42	130.04	0.9986	6.8825	3.6188	23.0059	0.0038
		Q6_K	130.42	130.50	0.9956	12.2018	11.3344	47.7428	0.0205
		Q5_K_M	130.42	130.31	0.9925	16.0152	11.3250	57.6408	0.0273
		Q4_K_M	130.42	130.73	0.9802	25.9705	11.3685	83.5772	0.0121
		Q3_K_M	130.42	133.80	0.9452	43.8480	11.6260	130.4067	0.1651
	BnB	Q2_K	130.42	134.51	0.8355	76.0838	11.6000	214.1535	0.4977
		INT8	130.42	131.30	0.9871	21.0565	0	54.7277	0.0277
		NF4	130.42	133.20	0.9490	42.1937	0	109.6652	0.1732
		FP4	130.42	135.19	0.8903	62.3800	0	162.1313	0.2850
	GPTQ	GPTQ-4bit	130.42	132.77	0.9407	45.3769	0	117.9387	0.0075
SQuAD ($r_s=0.61$)	GGUF	FP16	133.98	133.95	1.0000	0.0269	0	0.0699	0.0015
		Q8_0	133.98	133.97	0.9976	9.3232	3.6869	29.4460	0.0162
		Q6_K	133.98	133.75	0.9935	15.2218	11.6331	56.0145	0.0584
		Q5_K_M	133.98	133.78	0.9896	19.3140	11.6363	66.6551	0.0677
		Q4_K_M	133.98	133.72	0.9743	30.3447	11.6278	95.3129	0.1332
		Q3_K_M	133.98	134.76	0.9376	47.4720	11.7025	139.9340	0.1084
	BnB	Q2_K	133.98	135.89	0.8134	82.4513	11.7798	230.9575	0.0205
		INT8	133.98	134.24	0.9829	24.8297	0	64.5345	0.0152
		NF4	133.98	134.33	0.9430	45.2788	0	117.6838	0.2994
		FP4	133.98	135.37	0.8972	61.0943	0	158.7895	0.2282

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
	GPTQ	GPTQ-4bit	133.98	137.60	0.9206	54.2122	0	140.9024	0.4021
TriviaQA ($r_s=0.97$)	–	FP16	143.70	143.66	0.9999	1.5028	0	3.9060	0.0008
		GGUF Q8_0	143.70	143.69	0.9991	6.0960	4.0109	21.5163	0.0069
		Q6_K	143.70	143.67	0.9953	13.8623	12.4300	53.6079	0.0073
		Q5_K_M	143.70	143.78	0.9909	19.4203	12.4401	68.0682	0.0379
		Q4_K_M	143.70	143.71	0.9724	33.7936	12.4329	105.4155	0.1682
		Q3_K_M	143.70	144.10	0.9186	58.0673	12.4635	168.5481	0.3294
		Q2_K	143.70	144.13	0.7515	101.4677	12.4675	281.3554	1.0967
	BnB	INT8	143.70	143.85	0.9799	28.8222	0	74.9114	0.1057
		NF4	143.70	143.85	0.9351	51.8082	0	134.6541	0.0959
		FP4	143.70	144.30	0.8696	73.5483	0	191.1589	0.6411
	GPTQ	GPTQ-4bit	143.70	144.14	0.9032	63.3118	0	164.5533	0.4587
GSM8K ($r_s=0.78$)	–	FP16	120.55	120.52	1.0000	0.0306	0	0.0797	0.0017
		GGUF Q8_0	120.55	120.49	1.0000	0.0580	3.3627	4.9062	0.0055
		Q6_K	120.55	120.32	1.0000	0.2255	10.6604	15.6622	0.0056
		Q5_K_M	120.55	120.22	1.0000	0.3270	10.6504	15.9119	0.0077
		Q4_K_M	120.55	120.29	1.0000	0.2623	10.6584	15.7552	0.0040
		Q3_K_M	120.55	121.18	1.0000	0.6352	10.7633	16.8727	0.0316
		Q2_K	120.55	122.39	0.8963	55.3507	10.8879	159.2594	0.0087
	BnB	INT8	120.55	120.83	1.0000	0.2819	0	0.7327	0.0081
		NF4	120.55	120.67	1.0000	0.1197	0	0.3110	0.0086
		FP4	120.55	122.26	0.9697	29.9164	0	77.7554	0.0210
	GPTQ	GPTQ-4bit	120.55	123.64	1.0000	3.0919	0	8.0360	0.0323
WikiText ($r_s=0.96$)	–	FP16	136.21	136.21	1.0000	0.0001	0	0.0002	0.0037
		GGUF Q8_0	136.21	136.20	1.0000	0.0128	3.8276	5.4464	0.0084
		Q6_K	136.21	136.05	1.0000	0.1625	11.9638	17.3416	0.0094
		Q5_K_M	136.21	136.10	1.0000	0.1106	11.9684	17.2134	0.0195
		Q4_K_M	136.21	136.02	1.0000	0.1934	11.9602	17.4168	0.0560
		Q3_K_M	136.21	136.25	0.9751	30.4290	11.9755	96.0237	0.1211
		Q2_K	136.21	135.16	0.8358	77.7726	11.8830	218.9432	0.3534
	BnB	INT8	136.21	136.32	1.0000	0.1055	0	0.2741	0.0059
		NF4	136.21	136.00	0.9805	26.8622	0	69.8174	0.1505
		FP4	136.21	136.52	0.9325	50.1154	0	130.2545	0.2247
	GPTQ	GPTQ-4bit	136.21	137.84	0.9528	42.1538	0	109.5617	0.2886
FineWeb-Edu ($r_s=0.75$)	–	FP16	106.92	106.96	1.0000	0.0423	0	0.1101	0.0004
		GGUF Q8_0	106.92	106.94	1.0000	0.0270	3.8381	5.4980	0.0027
		Q6_K	106.92	106.83	1.0000	0.0817	11.9556	17.1201	0.0002
		Q5_K_M	106.92	106.83	1.0000	0.0834	11.9553	17.1242	0.0011
		Q4_K_M	106.92	106.66	1.0000	0.2530	11.9317	17.5315	0.0115
		Q3_K_M	106.92	106.80	1.0000	0.1138	11.9474	17.1920	0.0562
		Q2_K	106.92	105.93	1.0000	0.9888	11.8310	19.3016	0.2031
	BnB	INT8	106.92	106.93	1.0000	0.0108	0	0.0279	0.0078
		NF4	106.92	106.76	1.0000	0.1584	0	0.4116	0.0474
		FP4	106.92	107.02	1.0000	0.1084	0	0.2816	0.0837
	GPTQ	GPTQ-4bit	106.92	107.55	1.0000	0.6367	0	1.6547	0.1425

Table 13: Geometric decomposition for **Llama-3.1-8B-Instruct** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.57$)	–	FP16	145.60	145.61	0.9995	4.7460	0	12.3411	0.0001
		GGUF Q8_0	145.60	145.61	0.9970	11.2201	4.0600	34.9174	0.0107
		Q6_K	145.60	145.68	0.9934	16.7180	12.6488	61.3598	0.0144
		Q5_K_M	145.60	145.69	0.9879	22.6319	12.6578	76.7505	0.0141

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
	BnB	Q4_K_M	145.60	145.17	0.9592	41.5516	12.5722	125.8263	0.0028
		Q3_K_M	145.60	143.71	0.8828	70.0655	12.4614	199.8141	0.0103
		Q2_K	145.60	145.45	0.7456	103.8004	12.5859	287.7111	0.4741
		INT8	145.60	145.49	0.9795	29.4451	0	76.5661	0.0199
		NF4	145.60	146.24	0.9204	58.2411	0	151.4443	0.0432
		FP4	145.60	146.18	0.8657	75.6203	0	196.6353	0.0798
		GPTQ	145.60	145.43	0.8796	71.4043	0	185.6726	0.0119
MMLU ($r_s=0.74$)	GGUF	FP16	144.62	144.62	0.9997	3.4830	0	9.0568	0.0006
		Q8_0	144.62	144.53	0.9954	13.8093	4.0153	41.5868	0.0225
		Q6_K	144.62	144.62	0.9923	17.9276	12.4803	64.2670	0.0147
		Q5_K_M	144.62	144.62	0.9851	25.0044	12.4869	82.6780	0.0134
	BnB	Q4_K_M	144.62	143.11	0.9401	49.8141	12.2952	146.9196	0.1047
		Q3_K_M	144.62	141.90	0.8548	77.2496	12.1988	218.1237	0.0306
		Q2_K	144.62	144.24	0.7331	105.5181	12.4545	291.9918	0.6372
		INT8	144.62	144.44	0.9771	30.9202	0	80.4019	0.0423
		NF4	144.62	145.41	0.9155	59.6282	0	155.0510	0.1038
		FP4	144.62	144.77	0.8618	76.0598	0	197.7782	0.0551
		GPTQ	144.62	143.76	0.8708	73.3123	0	190.6339	0.0867
SQuAD ($r_s=0.97$)	GGUF	FP16	138.74	138.71	1.0000	0.0266	0	0.0693	0.0002
		Q8_0	138.74	138.76	0.9991	5.7734	3.7822	20.3615	0.0031
		Q6_K	138.74	138.79	0.9976	9.5680	11.8646	41.6587	0.0171
		Q5_K_M	138.74	138.77	0.9953	13.3976	11.8624	51.6138	0.0095
		Q4_K_M	138.74	138.89	0.9873	22.1163	11.8750	74.3028	0.0198
		Q3_K_M	138.74	138.92	0.9672	35.5472	11.8788	109.2326	0.0664
		Q2_K	138.74	138.30	0.8992	62.1981	11.8234	178.4544	0.3098
		INT8	138.74	138.72	0.9921	17.4038	0	45.2550	0.0130
	BnB	NF4	138.74	138.83	0.9693	34.3691	0	89.3699	0.0479
		FP4	138.74	139.36	0.9459	45.7438	0	118.9474	0.1904
		GPTQ	138.74	138.89	0.9665	35.9126	0	93.3836	0.0678
TriviaQA ($r_s=0.95$)	GGUF	FP16	143.91	143.91	1.0000	0.0014	0	0.0037	0.0010
		Q8_0	143.91	143.95	0.9994	4.8796	3.9966	18.3405	0.0021
		Q6_K	143.91	144.01	0.9969	11.3934	12.3712	47.1219	0.0210
		Q5_K_M	143.91	143.91	0.9926	17.5322	12.3603	63.0692	0.0654
		Q4_K_M	143.91	144.00	0.9781	30.1318	12.3735	95.8505	0.0653
		Q3_K_M	143.91	144.32	0.9288	54.3893	12.4065	158.9739	0.3116
		Q2_K	143.91	145.04	0.7730	97.3432	12.4896	270.7843	1.0636
		INT8	143.91	144.33	0.9843	25.5436	0	66.4209	0.0496
	BnB	NF4	143.91	144.84	0.9450	47.8877	0	124.5222	0.1588
		FP4	143.91	144.93	0.9044	63.1491	0	164.2065	0.3070
		GPTQ	143.91	144.13	0.9331	52.6843	0	136.9950	0.1271
GSM8K ($r_s=0.81$)	GGUF	FP16	134.52	134.54	1.0000	0.0210	0	0.0546	0.0004
		Q8_0	134.52	134.54	1.0000	0.0244	3.7839	5.4146	0.0007
		Q6_K	134.52	134.48	1.0000	0.0348	11.8717	16.8796	0.0019
		Q5_K_M	134.52	134.56	1.0000	0.0382	11.8793	16.8992	0.0087
		Q4_K_M	134.52	134.22	1.0000	0.3006	11.8493	17.5391	0.0028
		Q3_K_M	134.52	134.67	1.0000	0.1505	11.8831	17.1965	0.0215
		Q2_K	134.52	133.96	0.9129	56.0408	11.8313	162.4548	0.1025
		INT8	134.52	134.54	1.0000	0.0234	0	0.0608	0.0046
	BnB	NF4	134.52	134.43	1.0000	0.0866	0	0.2252	0.0274
		FP4	134.52	134.92	0.9815	25.9389	0	67.4488	0.0217
		GPTQ	134.52	134.95	1.0000	0.4297	0	1.1173	0.0302
WikiText ($r_s=0.63$)	GGUF	FP16	137.90	137.93	1.0000	0.0288	0	0.0749	0.0057
		Q8_0	137.90	137.92	1.0000	0.0152	3.8539	5.4896	0.0030
		Q6_K	137.90	137.85	1.0000	0.0539	12.0334	17.1580	0.0184
		Q5_K_M	137.90	137.87	1.0000	0.0307	12.0364	17.1020	0.0239

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
FineWeb-Edu ($r_s=0.79$)	BnB	Q4_K_M	137.90	137.73	1.0000	0.1780	12.0214	17.4635	0.0464
		Q3_K_M	137.90	137.55	1.0000	0.3545	12.0043	17.8983	0.0041
		Q2_K	137.90	136.60	0.9257	52.9351	11.9120	154.4932	0.0786
		INT8	137.90	137.89	1.0000	0.0135	0	0.0352	0.0055
		NF4	137.90	137.79	1.0000	0.1087	0	0.2826	0.0232
		FP4	137.90	137.98	0.9879	21.4694	0	55.8268	0.0698
	GPTQ	GPTQ-4bit	137.90	137.84	1.0000	0.0662	0	0.1721	0.0766
	GGUF	FP16	109.04	109.10	1.0000	0.0592	0	0.1539	0.0006
		Q8_0	109.04	109.05	1.0000	0.0126	3.8450	5.4703	0.0001
		Q6_K	109.04	109.04	1.0000	0.0017	12.0189	17.0017	0.0020
		Q5_K_M	109.04	109.05	1.0000	0.0157	12.0232	17.0442	0.0048
		Q4_K_M	109.04	109.01	1.0000	0.0331	12.0111	17.0724	0.0156
		Q3_K_M	109.04	108.93	1.0000	0.1124	11.9890	17.2471	0.0528
		Q2_K	109.04	108.25	1.0000	0.7898	11.8411	18.7997	0.2492
		INT8	109.04	109.01	1.0000	0.0294	0	0.0764	0.0094
		NF4	109.04	108.97	1.0000	0.0694	0	0.1804	0.0423
		FP4	109.04	108.97	1.0000	0.0680	0	0.1769	0.0854
		GPTQ	GPTQ-4bit	109.04	108.99	1.0000	0.0529	0	0.1375

Table 14: Geometric decomposition for **Ministral-3-8B-Instruct** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC ($r_s=0.87$)	GGUF	FP16	189.39	189.39	0.9999	2.2720	0	2.1085	0.0000
		Q8_0	189.39	189.32	0.9968	15.1145	2.7992	17.9855	0.0026
		Q6_K	189.39	189.18	0.9958	17.3340	8.6647	28.3403	0.0127
		Q5_K_M	189.39	189.42	0.9934	21.6905	8.6698	32.3906	0.0200
		Q4_K_M	189.39	189.59	0.9838	34.1040	8.6677	43.9077	0.0059
		Q3_K_M	189.39	187.25	0.9604	53.0135	8.6013	61.3626	0.0089
	BnB	Q2_K	189.39	187.79	0.8818	91.6906	8.5800	97.2264	0.1166
		INT8	189.39	189.26	0.9953	18.3774	0	17.0549	0.0028
		NF4	189.39	189.84	0.9609	53.0220	0	49.2064	0.0368
		FP4	189.39	188.99	0.9398	65.6746	0	60.9485	0.0762
MMLU ($r_s=0.73$)	GGUF	FP16	189.50	189.50	0.9999	2.3165	0	2.1498	0.0000
		Q8_0	189.50	189.43	0.9968	15.0827	2.8145	17.9777	0.0037
		Q6_K	189.50	189.44	0.9959	17.1656	8.7209	28.2635	0.0170
		Q5_K_M	189.50	189.78	0.9936	21.5038	8.7252	32.2957	0.0267
		Q4_K_M	189.50	189.47	0.9835	34.4325	8.7249	44.2935	0.0119
		Q3_K_M	189.50	187.06	0.9625	51.6260	8.6486	60.1419	0.0236
	BnB	Q2_K	189.50	187.52	0.8842	90.7349	8.6643	96.4587	0.1117
		INT8	189.50	189.33	0.9952	18.4668	0	17.1379	0.0101
		NF4	189.50	189.63	0.9625	51.9359	0	48.1985	0.0066
		FP4	189.50	189.18	0.9441	63.3063	0	58.7507	0.0564
SQuAD ($r_s=0.76$)	GGUF	FP16	184.40	184.40	1.0000	0.0000	0	0.0000	0.0000
		Q8_0	184.40	184.36	0.9973	13.4530	2.6285	16.2022	0.0259
		Q6_K	184.40	184.18	0.9966	15.2881	8.2666	25.8787	0.0033
		Q5_K_M	184.40	184.34	0.9945	19.3264	8.2698	29.6309	0.0275
		Q4_K_M	184.40	184.38	0.9869	29.8964	8.2676	39.4372	0.0112
		Q3_K_M	184.40	183.81	0.9651	48.6141	8.2588	56.7954	0.0239
	BnB	Q2_K	184.40	184.73	0.8963	84.0538	8.2634	89.6914	0.1217
		INT8	184.40	184.46	0.9965	15.5005	0	14.3850	0.0187
		NF4	184.40	184.56	0.9714	44.1227	0	40.9475	0.0661
		FP4	184.40	185.74	0.9524	57.1469	0	53.0345	0.0804
TriviaQA	–	FP16	193.86	193.86	1.0000	0.0000	0	0.0000	0.0000

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
(r _s =0.99)	GGUF	Q8_0	193.86	193.78	0.9968	15.5638	2.7147	18.2830	0.0083
		Q6_K	193.86	193.70	0.9956	18.1951	8.4494	28.8350	0.0032
		Q5_K_M	193.86	193.82	0.9932	22.6391	8.4506	32.9609	0.0264
		Q4_K_M	193.86	193.86	0.9835	35.1808	8.4491	44.5980	0.0389
		Q3_K_M	193.86	193.84	0.9510	60.6620	8.4503	68.2472	0.1490
		Q2_K	193.86	194.74	0.8363	111.1686	8.4679	115.1442	0.5791
	BnB	INT8	193.86	193.95	0.9945	20.3606	0	18.8954	0.0204
		NF4	193.86	194.07	0.9634	52.4936	0	48.7160	0.0883
		FP4	193.86	194.53	0.9354	69.8035	0	64.7803	0.1946
GSM8K (r _s =0.87)	GGUF	FP16	175.51	175.51	1.0000	0.0000	0	0.0000	0.0000
		Q8_0	175.51	175.44	1.0000	0.0676	2.5797	3.7110	0.0025
		Q6_K	175.51	175.32	1.0000	0.1925	8.1891	11.7597	0.0015
		Q5_K_M	175.51	175.29	1.0000	0.2167	8.1901	11.7837	0.0032
		Q4_K_M	175.51	175.40	1.0000	0.1080	8.1912	11.6843	0.0005
		Q3_K_M	175.51	174.95	1.0000	0.5605	8.1803	12.0889	0.0100
	BnB	Q2_K	175.51	176.60	0.9848	30.7295	8.2059	40.1231	0.0266
		INT8	175.51	175.55	1.0000	0.0452	0	0.0420	0.0002
		NF4	175.51	175.39	1.0000	0.1153	0	0.1070	0.0001
		FP4	175.51	176.81	1.0000	1.2996	0	1.2061	0.0076
WikiText (r _s =0.90)	GGUF	FP16	185.23	185.23	1.0000	0.0000	0	0.0000	0.0000
		Q8_0	185.23	185.10	1.0000	0.1308	2.6513	3.8709	0.0705
		Q6_K	185.23	185.03	1.0000	0.2003	8.3414	11.9824	0.0173
		Q5_K_M	185.23	185.21	1.0000	0.0245	8.3756	11.8677	0.0170
		Q4_K_M	185.23	185.00	1.0000	0.2281	8.3230	11.9822	0.0606
		Q3_K_M	185.23	184.67	1.0000	0.5573	8.3157	12.2774	0.1470
	BnB	Q2_K	185.23	185.07	0.9407	63.7851	8.4160	71.0970	0.2120
		INT8	185.23	185.21	1.0000	0.0199	0	0.0185	0.0046
		NF4	185.23	185.41	1.0000	0.1801	0	0.1671	0.1457
		FP4	185.23	185.77	0.9991	7.9106	0	7.3414	0.1607
FineWeb-Edu (r _s =0.76)	GGUF	FP16	146.11	146.11	1.0000	0.0000	0	0.0000	0.0000
		Q8_0	146.11	146.11	1.0000	0.0056	2.6451	3.7460	0.0022
		Q6_K	146.11	146.02	1.0000	0.0932	8.3261	11.8613	0.0015
		Q5_K_M	146.11	146.09	1.0000	0.0234	8.3306	11.8030	0.0020
		Q4_K_M	146.11	146.05	1.0000	0.0579	8.3258	11.8281	0.0131
		Q3_K_M	146.11	145.78	1.0000	0.3355	8.3229	12.0818	0.0421
	BnB	Q2_K	146.11	145.71	1.0000	0.3989	8.3341	12.1564	0.1554
		INT8	146.11	146.11	1.0000	0.0042	0	0.0039	0.0038
		NF4	146.11	146.03	1.0000	0.0789	0	0.0732	0.0296
		FP4	146.11	146.27	1.0000	0.1557	0	0.1445	0.0538

Table 15: Geometric decomposition for **Qwen3-8B-Instruct** under identity alignment ($W=I$) — all task group. Each benchmark section reports Spearman’s $r_s(\delta, |\Delta\mathcal{R}|)$ across all quantization variants.

Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC (r _s =0.73)	GGUF	FP16	165.05	164.96	1.0000	0.0883	0	0.3012	0.0025
		Q8_0	165.05	165.00	0.9998	2.9624	8.8392	22.6013	0.0052
		Q6_K	165.05	165.43	0.9984	9.4418	28.4395	72.4126	0.0299
		Q5_K_M	165.05	165.57	0.9962	14.3634	28.5070	89.2891	0.0046
		Q4_K_M	165.05	163.96	0.9904	22.8760	28.1639	117.8288	0.0994
	BnB	INT8	165.05	167.48	0.9934	19.2691	0	65.7006	0.0290
		NF4	165.05	165.71	0.9776	34.9830	0	119.2793	0.0251
		FP4	165.05	166.88	0.9531	50.8643	0	173.4289	0.1047
	GPTQ	GPTQ-8bit	165.05	164.39	0.9998	3.0781	0	10.4952	0.0000
MMLU	–	FP16	159.12	159.07	1.0000	0.0476	0	0.1622	0.0045

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Dataset	Family	Method	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
(r _s =0.72)	GGUF	Q8_0	159.12	159.35	0.9997	3.5975	8.5121	24.3041	0.0013
		Q6_K	159.12	160.10	0.9979	10.4631	27.4813	74.5398	0.0219
		Q5_K_M	159.12	159.94	0.9952	15.5966	27.4944	92.0616	0.0075
		Q4_K_M	159.12	157.23	0.9888	23.7090	27.0322	119.0684	0.0812
	BnB	INT8	159.12	161.47	0.9927	19.5555	0	66.6772	0.0537
		NF4	159.12	161.29	0.9773	34.1801	0	116.5416	0.0214
		FP4	159.12	161.12	0.9465	52.4286	0	178.7626	0.0455
	GPTQ	GPTQ-8bit	159.12	158.63	0.9998	3.5519	0	12.1107	0.0027
SQuAD (r _s =0.88)	–	FP16	150.60	150.48	1.0000	0.1256	0	0.4282	0.0109
	GGUF	Q8_0	150.60	150.60	1.0000	0.0069	7.8699	11.1531	0.0041
		Q6_K	150.60	149.92	0.9989	7.1005	24.9185	59.4501	0.0122
		Q5_K_M	150.60	151.20	0.9954	14.4602	25.1523	84.8748	0.1442
		Q4_K_M	150.60	150.24	0.9875	23.7815	24.9625	116.3885	0.1393
	BnB	INT8	150.60	150.93	0.9928	18.0966	0	61.7027	0.0660
		NF4	150.60	155.57	0.9656	40.4727	0	137.9971	0.2196
		FP4	150.60	154.65	0.9500	48.4088	0	165.0564	0.1680
	GPTQ	GPTQ-8bit	150.60	150.59	1.0000	0.0103	0	0.0350	0.0143
TriviaQA (r _s =0.62)	–	FP16	151.01	150.98	1.0000	0.0214	0	0.0731	0.0102
	GGUF	Q8_0	151.01	151.12	0.9999	2.4011	7.8083	19.2294	0.0260
		Q6_K	151.01	150.65	0.9975	10.6129	24.6840	71.0945	0.0351
		Q5_K_M	151.01	151.41	0.9935	17.2703	24.7983	93.9557	0.0280
		Q4_K_M	151.01	149.70	0.9820	28.5797	24.5577	132.1764	0.0152
	BnB	INT8	151.01	151.53	0.9907	20.6894	0	70.5433	0.2055
		NF4	151.01	153.15	0.9571	44.6209	0	152.1411	0.5815
		FP4	151.01	151.49	0.9338	55.0382	0	187.6604	0.4596
	GPTQ	GPTQ-8bit	151.01	150.97	0.9999	2.3989	0	8.1793	0.0421
GSM8K (r _s = − 0.53)	–	FP16	131.00	130.96	1.0000	0.0359	0	0.1223	0.0043
	GGUF	Q8_0	131.00	131.00	1.0000	0.0026	7.0929	10.0397	0.0063
		Q6_K	131.00	130.47	1.0000	0.5333	22.6102	33.7938	0.0001
		Q5_K_M	131.00	131.35	1.0000	0.3531	22.7810	33.4210	0.0043
		Q4_K_M	131.00	128.35	1.0000	2.6486	22.3022	40.5710	0.0001
	BnB	INT8	131.00	131.35	1.0000	0.3535	0	1.2053	0.0008
		NF4	131.00	132.32	1.0000	1.3216	0	4.5062	0.0119
		FP4	131.00	138.99	1.0000	7.9887	0	27.2384	0.0006
	GPTQ	GPTQ-8bit	131.00	130.86	1.0000	0.1356	0	0.4622	0.0017
WikiText (r _s =0.82)	–	FP16	149.22	149.22	1.0000	0.0025	0	0.0087	0.0027
	GGUF	Q8_0	149.22	149.14	1.0000	0.0788	8.0759	11.6899	0.0061
		Q6_K	149.22	149.15	1.0000	0.0659	25.6988	36.5682	0.0078
		Q5_K_M	149.22	149.33	1.0000	0.1110	25.7409	36.7816	0.0142
		Q4_K_M	149.22	148.02	1.0000	1.1917	25.5566	40.2056	0.1436
	BnB	INT8	149.22	149.33	1.0000	0.1096	0	0.3738	0.0027
		NF4	149.22	151.13	1.0000	1.9104	0	6.5138	0.1816
		FP4	149.22	155.58	1.0000	6.3667	0	21.7081	0.2169
	GPTQ	GPTQ-8bit	149.22	149.17	1.0000	0.0457	0	0.1559	0.0028
FineWeb-Edu (r _s =0.83)	–	FP16	136.78	136.83	1.0000	0.0427	0	0.1456	0.0009
	GGUF	Q8_0	136.78	136.61	1.0000	0.1781	7.8758	11.7455	0.0017
		Q6_K	136.78	136.95	1.0000	0.1661	25.0845	36.0413	0.0053
		Q5_K_M	136.78	136.86	1.0000	0.0777	25.0678	35.7161	0.0135
		Q4_K_M	136.78	135.47	1.0000	1.3156	24.8293	39.5995	0.0293
	BnB	INT8	136.78	136.68	1.0000	0.1042	0	0.3554	0.0049
		NF4	136.78	139.04	1.0000	2.2581	0	7.6992	0.0307
		FP4	136.78	143.35	1.0000	6.5636	0	22.3796	0.0684
	GPTQ	GPTQ-8bit	136.78	136.72	1.0000	0.0652	0	0.2225	0.0004

734 G Qwen3-8B Forgetting and Shape-Regularization Replications

735 This appendix reports the Qwen3-8B replications of Sec. 5.2 and Sec. 5.4 along with the remaining
 736 trace-norm decomposition tables (Llama BBQ and both Qwen fine-tuning tasks). Qwen3-8B differs
 737 from Llama-3.1-8B in depth (36 vs. 32 layers), vocabulary size (151,936 vs. 128,256), tokenizer
 738 family (byte-level BPE vs. BPE), and pre-training corpus (multilingual, ~ 36 T tokens vs. English-
 739 dominant, ~ 15 T tokens); reproducing the main-text patterns under these shifts provides cross-family
 740 evidence for the claims of Secs. 5.2–5.4.

741 G.1 Forgetting: Bound Tracks Cross-Task Drift on Qwen3-8B

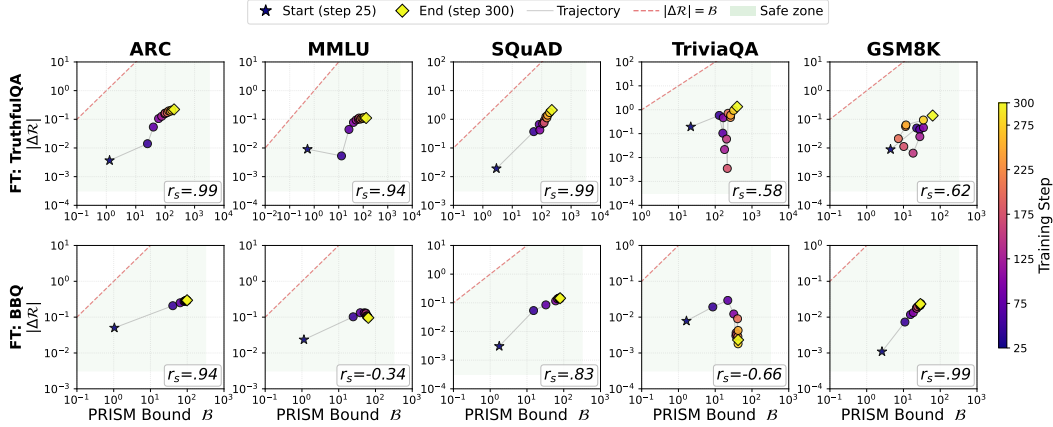


Figure 7: **Qwen3-8B: the PRISM bound tracks catastrophic forgetting across LoRA fine-tuning steps.** Each subplot scatters the PRISM bound B (x-axis, log) against the empirical forgetting $|\Delta\mathcal{R}|$ (y-axis, log) on a held-out benchmark, with one point per LoRA checkpoint colored by training step. Rows: fine-tuning task (TruthfulQA, BBQ). Columns: held-out evaluation benchmark (ARC, MMLU, SQuAD, TriviaQA, GSM8K). Because LoRA keeps the `lm_head` frozen, the head-discrepancy term γ vanishes and the bound is governed entirely by backbone scale ($\Delta\rho$) and shape ($1 - \Omega$) drift, as predicted by Sec. 4. Spearman r_s per subplot is annotated in each panel. This replicates Fig. 3 on a different family, whose pre-training corpus, depth, vocabulary size, and tokenizer family all differ from Llama-3.1-8B.

742 Figure 7 reproduces the two structural observations of the Llama main-text result (Fig. 3) on
 743 Qwen3-8B. First, *the PRISM bound rises in lockstep with empirical forgetting across most*
 744 *(task, benchmark) combinations*: as LoRA fine-tuning proceeds on TruthfulQA or BBQ, $\Omega(Z_0, Z_t)$
 745 drifts away from 1 and ρ_t from ρ_0 , and this backbone drift tracks the empirical cross-entropy drift
 746 on most held-out benchmarks—including benchmarks (e.g., GSM8K) whose distributions differ
 747 substantially from the fine-tuning data. Second, *different source tasks induce qualitatively different*
 748 *drift geometries*, matching Llama: TruthfulQA fine-tuning drives predominantly shape drift while
 749 BBQ moves both scale and shape, a distinction the decomposition of Eq. (8) makes visible whereas a
 750 unified distance would collapse it. Because Qwen3-8B differs from Llama-3.1-8B in every one of
 751 the four variables held out in Sec. 5.1 (pre-training corpus, depth, vocabulary, tokenizer family), this
 752 replication is direct evidence that the backbone-drift signal is a cross-family property rather than a
 753 Llama-specific artifact.

754 **Per-subplot variability under low-forgetting regimes.** A small subset of subplots in Fig. 7—most
 755 notably Qwen3-8B fine-tuned on BBQ evaluated on MMLU and TriviaQA—shows weak or slightly
 756 negative per-subplot r_s . These cases share a common signature: the baseline empirical forgetting
 757 at $\lambda=0$ is at the noise floor (e.g., Qwen3-8B BBQ on TriviaQA: $|\Delta\mathcal{R}|=0.0023$ at $\lambda=0$, vs. Llama
 758 TruthfulQA on TriviaQA: $|\Delta\mathcal{R}|=4.63$ —three orders of magnitude apart). When the forgetting
 759 magnitude itself is below per-evaluation noise, per-checkpoint ranking becomes uninformative; the
 760 resulting r_s reflects evaluation noise rather than a failure of the bound. This pattern is a property of
 761 the model–task pairing (Qwen3-8B is robust to BBQ fine-tuning) rather than the diagnostic.

Table 16: Trace-norm forgetting decomposition for **Llama-3.1-8B** fine-tuned on **BBQ** under identity alignment ($W=I$). Rows group by evaluation benchmark; each benchmark lists metrics at step 300 for $\lambda \in \{0.0, 0.1, 0.5\}$ (target = base model). Within each benchmark the row with the smallest $|\Delta\mathcal{R}|$ is marked: both $|\Delta\mathcal{R}|$ and the corresponding Ω are in bold.

Dataset	λ	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC	$\lambda=0.0$	137.55	135.31	0.8762	67.9234	0	67.9234	0.1884
	$\lambda=0.1$	137.55	135.71	0.9570	40.1190	0	104.8606	0.1846
	$\lambda=0.5$	137.55	136.75	0.9732	31.7369	0	82.9521	0.1892
MMLU	$\lambda=0.0$	138.91	136.48	0.8759	68.6286	0	68.6286	0.5218
	$\lambda=0.1$	138.91	137.20	0.9563	40.8522	0	106.7771	0.6239
	$\lambda=0.5$	138.91	138.28	0.9724	32.5811	0	85.1586	0.5441
SQuAD	$\lambda=0.0$	138.20	138.84	0.9429	46.8184	0	46.8184	0.2117
	$\lambda=0.1$	138.20	138.82	0.9508	43.4465	0	113.5579	0.4110
	$\lambda=0.5$	138.20	138.53	0.9633	37.4890	0	97.9867	0.3216
TriviaQA	$\lambda=0.0$	140.93	141.51	0.9841	25.2080	0	25.2080	0.1083
	$\lambda=0.1$	140.93	141.38	0.9892	20.7730	0	54.2951	0.0539
	$\lambda=0.5$	140.93	141.13	0.9945	14.7452	0	38.5401	0.0161
GSM8K	$\lambda=0.0$	137.15	137.36	1.0000	0.2161	0	0.2161	0.0478
	$\lambda=0.1$	137.15	137.41	1.0000	0.2669	0	0.6977	0.0300
	$\lambda=0.5$	137.15	137.22	1.0000	0.0778	0	0.2034	0.0152

762 G.2 Shape Regularization: Qwen3-8B Replication

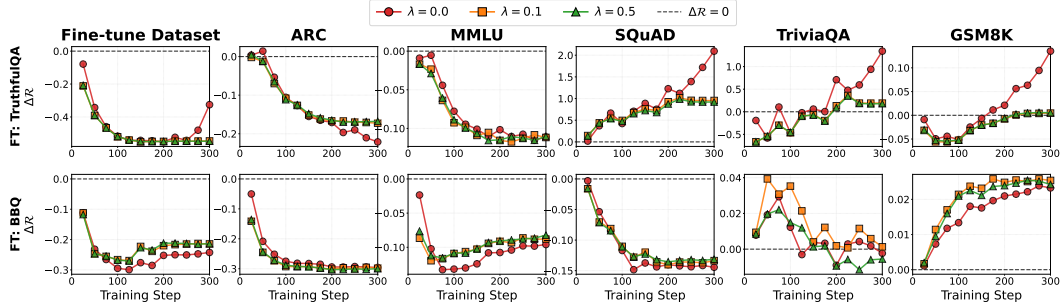


Figure 8: **PRISM-guided shape regularization on Qwen3-8B (replication of Fig. 4)**. LoRA fine-tuning on TruthfulQA (top row) and BBQ (bottom row) with the auxiliary penalty $\lambda(1 - \Omega(Z_0^{\text{ref}}, Z_t^{\text{ref}}))$ at $\lambda \in \{0.0, 0.1, 0.5\}$ and analysis step 300. The uniform downstream-forgetting reduction and the monotone trade-off between target-task fit and downstream damage seen on Llama hold on Qwen3-8B as well, confirming that the shape regularizer of Sec. 3.5 is not family-specific.

Figure 8 reproduces the main-text Llama result (Fig. 4) on Qwen3-8B: $\lambda \in \{0.0, 0.1, 0.5\}$ produces the same monotone reduction in downstream $|\Delta\mathcal{R}|$ across all five benchmarks for both TruthfulQA and BBQ fine-tuning, confirming the regularizer’s effect is family-agnostic. Per-variant numerical decompositions are in Tables 17 and 18 below.

767 G.3 Additional Trace-Norm Decomposition Tables

Tables 16–18 give the full per-(model, fine-tuning task) decomposition at step 300 for $\lambda \in \{0.0, 0.1, 0.5\}$. The main text (Table 2) showed the Llama-TruthfulQA combination; this appendix completes the 2×2 grid.

Table 17: Trace-norm forgetting decomposition for **Qwen3-8B-Base** fine-tuned on **TruthfulQA** under identity alignment ($W=I$). Rows group by evaluation benchmark; each benchmark lists metrics at step 300 for $\lambda \in \{0.0, 0.1, 0.5\}$ (target = base model). Within each benchmark the row with the smallest $|\Delta\mathcal{R}|$ is marked: both $|\Delta\mathcal{R}|$ and the corresponding Ω are in bold.

Dataset	λ	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC	$\lambda=0.0$	332.98	316.08	0.9854	58.0426	0	58.0426	0.2211
	$\lambda=0.1$	332.98	326.97	0.9958	30.8029	0	106.5250	0.1727
	$\lambda=0.5$	332.98	327.34	0.9958	30.8393	0	106.6508	0.1678
MMLU	$\lambda=0.0$	333.53	323.66	0.9938	37.9230	0	37.9230	0.1110
	$\lambda=0.1$	333.53	328.73	0.9983	19.9632	0	69.0384	0.1117
	$\lambda=0.5$	333.53	328.95	0.9983	19.8459	0	68.6326	0.1103
SQuAD	$\lambda=0.0$	297.68	277.66	0.9777	63.9596	0	63.9596	2.0940
	$\lambda=0.1$	297.68	292.49	0.9927	36.0801	0	124.7748	0.9542
	$\lambda=0.5$	297.68	292.84	0.9930	35.3768	0	122.3427	0.9197
TriviaQA	$\lambda=0.0$	240.42	201.68	0.8815	113.9728	0	113.9728	1.3422
	$\lambda=0.1$	240.42	224.47	0.9655	63.0973	0	218.2079	0.1921
	$\lambda=0.5$	240.42	224.99	0.9661	62.4935	0	216.1197	0.1794
GSM8K	$\lambda=0.0$	269.06	251.07	1.0000	17.9976	0	17.9976	0.1352
	$\lambda=0.1$	269.06	272.59	1.0000	3.5224	0	12.1813	0.0047
	$\lambda=0.5$	269.06	272.33	1.0000	3.2676	0	11.3001	0.0055

Table 18: Trace-norm forgetting decomposition for **Qwen3-8B-Base** fine-tuned on **BBQ** under identity alignment ($W=I$). Rows group by evaluation benchmark; each benchmark lists metrics at step 300 for $\lambda \in \{0.0, 0.1, 0.5\}$ (target = base model). Within each benchmark the row with the smallest $|\Delta\mathcal{R}|$ is marked: both $|\Delta\mathcal{R}|$ and the corresponding Ω are in bold.

Dataset	λ	ρ_T	ρ_P	Ω	δ	γ	$\mathcal{B}$	$ \Delta\mathcal{R} $
ARC	$\lambda=0.0$	332.98	323.65	0.9966	28.6389	0	28.6389	0.2959
	$\lambda=0.1$	332.98	325.07	0.9977	23.6218	0	81.6908	0.2983
	$\lambda=0.5$	332.98	325.89	0.9983	20.4425	0	70.6960	0.2999
MMLU	$\lambda=0.0$	333.53	327.67	0.9986	18.4691	0	18.4691	0.0959
	$\lambda=0.1$	333.53	328.63	0.9990	15.5053	0	53.6217	0.0879
	$\lambda=0.5$	333.53	329.21	0.9990	15.3953	0	53.2413	0.0822
SQuAD	$\lambda=0.0$	297.68	285.26	0.9978	23.0185	0	23.0185	0.1444
	$\lambda=0.1$	297.68	286.07	0.9985	19.7367	0	68.2550	0.1337
	$\lambda=0.5$	297.68	287.07	0.9990	16.9506	0	58.6198	0.1312
TriviaQA	$\lambda=0.0$	240.42	235.37	0.9989	12.1057	0	12.1057	0.0023
	$\lambda=0.1$	240.42	235.44	0.9991	11.1248	0	38.4726	0.0013
	$\lambda=0.5$	240.42	235.67	0.9992	10.5391	0	36.4471	0.0055
GSM8K	$\lambda=0.0$	269.06	260.69	1.0000	8.3704	0	8.3704	0.0233
	$\lambda=0.1$	269.06	261.31	1.0000	7.7529	0	26.8118	0.0254
	$\lambda=0.5$	269.06	262.16	1.0000	6.8995	0	23.8602	0.0242

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1112 Justification: The paper does not involve human subjects research; no IRB approval is
 1113 required.

1114 Guidelines:

- 1115 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
 1116 with human subjects.
- 1117 • Depending on the country in which research is conducted, IRB approval (or equivalent)
 1118 may be required for any human subjects research. If you obtained IRB approval, you
 1119 should clearly state this in the paper.
- 1120 • We recognize that the procedures for this may vary significantly between institutions
 1121 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
 1122 guidelines for their institution.
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 1124 applicable), such as the institution conducting the review.

1125 **16. Declaration of LLM usage**

1126 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 1127 non-standard component of the core methods in this research? Note that if the LLM is used
 1128 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 1129 scientific rigor, or originality of the research, declaration is not required.

1130 Answer: [N/A]

1131 Justification: LLMs are the *subject* of analysis in this work (Llama-3.1-8B, Qwen3-8B,
 1132 Ministral-3-8B, DeepSeek-R1-Distill-Llama-8B), not a tool used in developing the PRISM
 1133 methodology. No LLMs were used as an important, original, or non-standard component of
 1134 the core method.

1135 Guidelines:

- 1136 • The answer [N/A] means that the core method development in this research does not
 1137 involve LLMs as any important, original, or non-standard components.
- 1138 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
 1139 be described.